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DOCTORAL THESIS

High Density Rhythm Thesis for Stroke Rehabilitation in Piano

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in the

Research Group Name
Department or School Name

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Declaration of Authorship

I, Paul-Peter ARSLAN, declare that this thesis titled, "High Density Rhythm Thesis for Stroke Rehabilitation in Piano" and the work presented in it are my own. I confirm that:

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"Thanks to my solid academic training, today I can write hundreds of words on virtually any topic without possessing a shred of information, which is how I got a good job in journalism."

Dave Barry

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Abstract

Faculty Name
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Doctor of Philosophy

High Density Rhythm Thesis for Stroke Rehabilitation in Piano

by Paul-Peter ARSLAN

The Thesis Abstract is written here (and usually kept to just this page). The page is kept centered vertically so can expand into the blank space above the title too...

Acknowledgements

The acknowledgments and the people to thank go here, don't forget to include your project advisor…

Contents

Declaration of Authorship iii

Abstract vii

Acknowledgements ix

1 Introduction 1
1.1 Context & Clinical Need . . . 2
1.1.1 Stroke Burden and the Persistent Challenge of Upper-Limb Recovery . . . 2
1.1.2 The Neuromuscular and Sensorimotor Foundations of Dexterity Deficits After Stroke . . . 3
1.1.3 Limitations of Conventional Rehabilitation and the Need for Precision-Based, Technology-Augmented Approaches . . . 4
1.1.4 State of the Art: Measurement, Decoding, and Intervention Technologies for Fine Manual Dexterity . . . 8
Objective Measurement of Manual Dexterity: The Dextrain Manipulandum and Instrumented Assessment . . . 8
Neuromuscular Decoding and Myoelectric Control: From Surface EMG to High-Density Arrays and AI-Driven Regression . . . 9
Rhythm- and Music-Based Interventions: Evidence, Mechanisms, and Gaps . . . 11
Synthesis: Technology Landscape and Positioning of This Thesis 13
1.2 Why Music as a Precision Testbed? . . . 21
1.2.1 Why Music as a Precision Testbed? . . . 21
Structured Temporal Demands: Music as a Scaffold for Sensorimotor Synchronization . . . 21
Neuroscience Evidence Base, Clinical Promise, and the Measurement Gap . . . 24
Auditory Feedback as an Experimental and Clinical Lever: Probing Internal Control and Informing Design . . . 28
Synthesis and Transition: Music as the Ideal Convergence Point for the Tripartite Research Program . . . 31
1.3 Our Approach: Three Complementary Lenses . . . 33
1.3.1 Rhythm Karaoke (RK): The Measurement Lens—Validated Timing Precision Assessment for Fine Manual Dexterity . . . 34
1.3.2 HD-EMG + AI: The Mechanism Lens—Decoding Continuous Finger Forces from High-Density Neuromuscular Signals to Reveal Control Substrates . . . 35
1.3.3 ReTouche: The Engagement Lens—Defining Human-Centered Design Principles for Sustained Practice and Agency in Fine Motor Learning . . . 36

1.4 Research Questions, Hypotheses & Scope 37
1.4.1 Research Questions 37
1.4.2 Hypotheses 38
Hypothesis Set 1: Psychometric Validation of RK (Addressing RQ1) 38
Hypothesis Set 2: HD-EMG Decoding Performance and Mechanistic Linkage (Addressing RQ2) 38
Hypothesis Set 3: Engagement Design Principles from ReTouche (Addressing RQ3) 39
1.4.3 Scope and Limitations 39

A Frequently Asked Questions **41**
A.1 How do I change the colors of links? 41

Bibliography **43**

List of Figures

List of Tables

1.1 Measurement and Decoding: Strengths and Gaps. UL = upper limb; HD-EMG = high-density EMG; DOF = degrees of freedom; FMA = Fugl-Meyer Assessment; LoE = Level of Evidence. 16
1.2 Training and Feedback: Strengths and Gaps. UL = upper limb; RAS = rhythmic auditory stimulation; MST = music-supported therapy; KR/KP = knowledge of results/performance; RCT = randomized controlled trial; SR = systematic review; LoE = Level of Evidence. 19
1.3 The three projects of this thesis, their objectives, methods with quantitative specifications, gap-filling contributions, and status. 33

List of Abbreviations

LAH	**L**ist **A**bbreviations **H**ere
WSF	**W**hat (it) **S**tands **F**or

For/Dedicated to/To my…

Chapter 1

Introduction

Purpose & Stance

Stroke remains a leading cause of long-term disability, affecting approximately 15 million people annually (Feigin et al., 2021). While gross motor recovery has received substantial clinical attention, the restoration of *fine manual dexterity*—the precise coordination of timing, force, and finger individuation—represents a more challenging frontier for functional independence (Kwakkel et al., 2003; Voet et al., 2019; Keh et al., 2022). Post-stroke survivors exhibit profound neuromuscular disorganization: elevated co-contraction, impaired motor unit recruitment, degraded sensorimotor integration, and compromised timing precision (Kamper et al., 2006; De Luca et al., 2009; Repp and Su, 2013). Traditional rehabilitation approaches yield modest gains, with persistent challenges in measuring fine-grained improvements, understanding neuromuscular mechanisms, and sustaining long-term engagement (Kwakkel et al., 2008; Weegen et al., 2016).

The central purpose of this thesis is to investigate *precision in hand dexterity after stroke* through a coherent framework integrating three complementary lenses: **measurement** (quantifying timing precision with validated music-based protocols), **mechanism** (decoding neuromuscular substrates via high-density electromyography and artificial intelligence), and **engagement** (understanding lived experience through human-computer interaction inquiry). We leverage *music* as a structured, ecologically valid testbed that imposes tight temporal constraints, engages auditory-motor networks, and provides intrinsic motivation (Patel, 2008; Thaut et al., 1999; Miendlarzewska and Trost, 2014). This tripartite architecture positions music not as motivational overlay but as a *quantitative precision testbed* bridging laboratory rigor and real-world rehabilitation practice. The thesis adopts a *mixed-methods ethos*: psychometric validation, machine learning benchmarks, and mechanistic modeling are complemented by qualitative inquiry (autoethnography, focus groups) to address the biological, behavioral, technological, and experiential dimensions of neurorehabilitation. By clearly delineating scope—measurement tool development, proof-of-concept mechanistic exploration, and design-oriented inquiry rather than randomized efficacy trials or clinical deployment—we position this work as foundational for future translational research in precision-based, music-augmented, user-centered rehabilitation of hand dexterity after stroke.

1.1 Context & Clinical Need

1.1.1 Stroke Burden and the Persistent Challenge of Upper-Limb Recovery

Stroke remains one of the leading causes of long-term disability worldwide, imposing an enormous burden on individuals, families, and healthcare systems. Each year, approximately 15 million people suffer a stroke globally, with roughly one-third dying and another one-third left with permanent disability (Feigin et al., 2021). In the United States alone, stroke affects nearly 800,000 individuals annually and is responsible for substantial morbidity, with an estimated 7 million stroke survivors living with residual impairments (Tsao et al., 2023). Among the constellation of deficits following stroke—including hemiparesis, sensory loss, aphasia, and cognitive impairment—upper-limb (UL) motor dysfunction is among the most prevalent and debilitating, affecting approximately 80% of acute stroke patients and persisting in 50-60% of survivors beyond six months post-onset (Kwakkel et al., 2003; Parker, Wade, and Langton Hewer, 1986). This persistent upper-limb impairment directly compromises functional independence and quality of life, limiting the ability to perform activities of daily living such as dressing, eating, grooming, writing, and manipulating objects, which collectively depend on coordinated, fine-grained control of the hand and fingers.

While significant clinical and research attention has historically focused on gross motor recovery—such as restoring reaching range of motion, regaining grip strength sufficient to hold objects, or achieving basic scores on ordinal functional scales like the Fugl-Meyer Assessment or Action Research Arm Test—the restoration of *fine manual dexterity* represents a more challenging and yet equally critical frontier for meaningful functional recovery (Voet et al., 2019; Kwakkel et al., 2003). Manual dexterity is not a unitary construct that can be adequately captured by simple strength measurements or binary success/failure on functional tasks. Rather, it encompasses a multidimensional set of control capabilities that enable the hand to perform skilled, precise, and adaptive manipulations: the ability to *time* finger movements accurately relative to external cues or internal rhythmic goals (with millisecond-level precision), to *scale forces* smoothly and proportionally across a wide dynamic range (from sub-Newton contacts required for delicate object manipulation to multi-Newton grips for stable holds), and to *individuate* digits (moving one finger independently while minimizing unwanted co-activation or overflow to other fingers) (Keh et al., 2022; De Luca et al., 2009). These three dimensions—timing precision, force scaling, and individuation—are foundational to virtually all skilled manual behaviors, from typing on a keyboard and buttoning a shirt, to playing a musical instrument and performing surgical tasks. They are also precisely the dimensions that are most frequently and severely compromised after stroke, reflecting fundamental disruptions in the neural control of movement.

The centrality of fine hand function to independence and participation in daily life cannot be overstated. Epidemiological studies consistently demonstrate that individuals who fail to recover sufficient hand dexterity experience markedly reduced quality of life, lower rates of return to work, increased caregiver burden, and higher long-term healthcare costs (Kwakkel et al., 2003; Parker, Wade, and Langton Hewer, 1986). Moreover, the deficits in manual dexterity are not merely a matter of lost strength or range; they reflect deeper impairments in the neural coordination and control processes that orchestrate complex, temporally precise, and force-modulated

finger movements. Traditional clinical outcome measures, such as grip strength dynamometry, the Nine-Hole Peg Test, or coarse ordinal scales, capture only a fraction of this complexity and are often insensitive to subtle yet functionally significant deficits in timing accuracy, force steadiness, or independent finger control (Keh et al., 2022). Consequently, there exists a pressing clinical and scientific need to develop *quantitative, high-resolution, multi-dimensional assessments* of manual dexterity that can reveal the precise nature of post-stroke deficits, inform the design of targeted interventions, and track recovery trajectories with sensitivity sufficient to detect meaningful but incremental improvements.

1.1.2 The Neuromuscular and Sensorimotor Foundations of Dexterity Deficits After Stroke

To understand why fine manual dexterity is so challenging to restore after stroke, it is essential to consider the underlying neuromuscular and sensorimotor mechanisms that support skilled finger control in the intact nervous system and how these mechanisms are disrupted by stroke. In healthy individuals, dexterous manipulation depends on the coordinated recruitment of multiple muscles spanning the forearm (extrinsic muscles that flex and extend the fingers) and the hand itself (intrinsic muscles including the thenar, hypothenar, interossei, and lumbricals that contribute to fine individuation and force modulation). This muscular control is orchestrated by descending corticospinal pathways that originate in the primary motor cortex (M1) and make direct, monosynaptic connections to spinal motor neurons innervating hand muscles—a unique anatomical feature in primates that enables fractionated, individuated finger movements (Cramer et al., 2011; Ward et al., 2003). Superimposed on this direct corticospinal control is a network of sensorimotor integration processes that incorporate proprioceptive, tactile, and auditory feedback to enable online error correction, adaptive force scaling, and anticipatory control based on internal models of limb dynamics and task demands (Wong et al., 2019).

Stroke disrupts these finely tuned systems at multiple levels. Lesions affecting the motor cortex, corticospinal tract, or subcortical structures (such as the basal ganglia and thalamus) compromise descending motor commands, leading to weakness (paresis), loss of selective motor control, and emergence of pathological synergies in which movements become stereotyped and coupled across multiple joints (Cramer et al., 2011; Ward et al., 2003). At the level of the motor unit and muscle, stroke survivors exhibit profound alterations in motor unit recruitment and discharge patterns: reduced motor unit firing rates, decreased recruitment of high-threshold motor units, increased variability (jitter) in discharge timing, and altered synchronization across motor units (Kamper et al., 2006). These changes manifest behaviorally as increased motor noise (variability in force output even during sustained contractions), difficulty in producing smooth, graded force increments (especially at low force levels critical for fine manipulation), and impaired ability to sustain forces steadily over time without drift or tremor (De Luca et al., 2009; Kamper et al., 2006).

Furthermore, stroke disrupts the selectivity of muscle activation, resulting in elevated co-contraction of agonist and antagonist muscle groups. While some degree of co-contraction is functionally adaptive for joint stabilization in healthy individuals, excessive and inappropriate co-contraction after stroke increases joint stiffness, reduces movement efficiency, elevates energy expenditure, and most critically, impairs finger individuation: attempts to move one finger independently often result in unwanted activation and movement of adjacent fingers, a phenomenon known as motor overflow (Keh et al., 2022). This deficit in individuation is not merely a

consequence of weakness but reflects a more fundamental loss of the fractionated corticospinal control that underlies independent digit movements. Studies using transcranial magnetic stimulation, neuroimaging, and fine-wire electromyography have shown that stroke survivors exhibit reduced cortical map specificity, increased cortical motor thresholds, reduced intracortical facilitation, and abnormal coupling between intended and actual muscle activations (Hummel and Cohen, 2006; Ward et al., 2003).

In addition to these motor control deficits, stroke compromises sensorimotor integration—the processes by which sensory feedback is used to guide and refine motor output. Proprioceptive impairments are common after stroke, reducing the accuracy of limb position sense, movement detection, and force perception (Wong et al., 2019). Tactile deficits impair the ability to detect contact, gauge object properties, and modulate grip forces in response to slips or load perturbations. Even when primary sensory modalities are relatively preserved, the integration of multisensory information and the construction of internal models for predictive control are often degraded (Wong et al., 2019). This sensorimotor integration deficit is particularly relevant for tasks that demand tight temporal coupling between sensory cues and motor output, such as synchronizing finger movements to auditory rhythms or adjusting force in response to haptic feedback. Experimental studies in stroke populations have documented increased asynchrony (temporal offset between cue and movement), increased variability (jitter) in synchronization, reduced ability to correct errors from one movement cycle to the next (impaired error correction or phase correction), and difficulty in maintaining stable rhythmic coordination over extended periods (phase drift) (Repp and Su, 2013; Schmidt and Lee, 2011). These timing deficits are not merely epiphenomena of weakness or slowness but reflect deeper impairments in the neural circuitry (involving cerebellum, basal ganglia, and premotor-motor cortical loops) that supports temporal prediction, internal pacing, and adaptive synchronization (Patel, 2008; Thaut et al., 1999).

Taken together, the neuromuscular and sensorimotor disruptions following stroke create a complex phenotype of dexterity impairment characterized by: (1) increased motor noise and reduced force steadiness, particularly at low force levels; (2) impaired force scaling and difficulty in producing smooth, graded force modulations; (3) reduced finger individuation with excessive co-contraction and motor overflow; (4) degraded timing precision and increased jitter in rhythmic or cued movements; and (5) compromised sensorimotor integration and adaptive error correction. These deficits cannot be adequately captured by traditional clinical scales or binary functional tests, nor can they be effectively targeted by rehabilitation interventions that treat hand function as a monolithic entity. Instead, a precision-based approach is required: one that measures these distinct dimensions at high resolution, understands their neuromuscular substrates through direct physiological monitoring, and designs interventions that address each dimension with targeted, feedback-enriched training protocols.

1.1.3 Limitations of Conventional Rehabilitation and the Need for Precision-Based, Technology-Augmented Approaches

Conventional post-stroke rehabilitation for the upper limb is grounded in well-established principles of motor learning and neuroplasticity: high-dose, task-specific, repetitive practice of functional movements delivered with appropriate feedback and progressive challenge (Kwakkel et al., 2008; Schmidt and Lee, 2011). Constraint-induced movement therapy (CIMT), for example, enforces use of the paretic limb

by restraining the non-paretic limb and prescribing intensive, structured practice of functional tasks; this approach has demonstrated efficacy in improving arm function in selected stroke survivors (Kwakkel et al., 2008). Similarly, task-oriented training focuses on practicing ecologically valid activities of daily living (reaching, grasping, manipulating objects) in contexts that mimic real-world demands. While these approaches yield measurable improvements in gross motor function and performance on clinical scales, their effects on fine manual dexterity—particularly the millisecond-level timing precision, sub-Newton force control, and independent finger individuation—remain modest and heterogeneous (Voet et al., 2019; Lee et al., 2021). Several factors contribute to this limitation.

First, conventional therapy typically lacks the *measurement precision* necessary to detect and target subtle deficits in the temporal, force, and individuation domains. Clinical assessments such as the Fugl-Meyer UL subscale, Action Research Arm Test, or Box and Block Test provide ordinal scores or count-based metrics that are useful for gross categorization of impairment severity but are insensitive to the fine-grained control variables that underlie skilled dexterity (Keh et al., 2022). For instance, a patient may achieve a "pass" on a grasp-and-release task by compensating with excessive force, prolonged movement times, or multi-finger synergies, yet the underlying deficits in force scaling, timing precision, and individuation remain unaddressed and unmeasured. Without high-resolution, continuous measures of these control dimensions, therapists cannot titrate feedback, adjust task difficulty, or track incremental progress with sufficient sensitivity.

Second, conventional therapy faces pervasive challenges with *dose and adherence*. Motor recovery and neuroplasticity are dose-dependent: meta-analyses and systematic reviews consistently demonstrate that greater amounts of practice (measured in hours per week) are associated with larger functional gains (Lee et al., 2021; Kwakkel et al., 2008). However, delivering the requisite high doses of practice is logistically and economically challenging in constrained healthcare systems, and adherence to home-based self-directed practice programs is notoriously poor, with dropout rates often exceeding 50% and actual practice time falling far short of prescribed dosages (Weegen et al., 2016; Verrienti et al., 2023). This adherence problem is partly motivational: repetitive, decontextualized exercises that lack intrinsic appeal or clear functional relevance fail to sustain engagement, particularly when practiced in isolation at home without therapist supervision or social support.

Third, conventional therapy provides limited *mechanistic insight* into *why* certain individuals recover better than others or *how* specific training parameters influence neural and muscular reorganization. Behavioral performance metrics (e.g., time to complete a task, number of successful trials) describe outcomes but do not reveal the underlying neuromuscular processes—motor unit recruitment patterns, muscle synergies, co-contraction dynamics, timing of neural drive—that support or constrain those outcomes. Without such mechanistic understanding, therapy remains empirical and generic rather than personalized and theoretically grounded.

In response to these limitations, the field has increasingly turned to *technology-assisted rehabilitation* approaches that promise to augment conventional therapy by providing high-intensity practice, objective performance metrics, engaging interfaces, and novel forms of feedback. These approaches span a wide spectrum, including robot-assisted therapy, virtual reality and gamified training environments, brain-computer interfaces (BCIs), functional electrical stimulation, movement sonification, and music- or rhythm-based interventions. Each modality brings unique strengths but also faces substantive limitations that have prevented widespread clinical adoption and durable functional impact.

Robot-assisted therapy uses robotic exoskeletons or end-effector devices to guide, assist, or resist upper-limb movements, enabling high-dose, precisely controlled practice with real-time kinematic feedback. Systematic reviews and meta-analyses have shown that robot-assisted training can improve motor function as measured by clinical scales, particularly in individuals with moderate to severe impairment (Kwakkel et al., 2008; Burdea et al., 2019). However, effect sizes are generally modest, benefits often fail to generalize beyond the trained movements or robotic environment, and there is limited evidence that robot-assisted therapy improves fine finger dexterity or real-world functional independence more effectively than conventional dose-matched therapy (Kwakkel et al., 2008). Moreover, robotic systems are expensive, cumbersome, and typically confined to clinical or research settings, limiting accessibility and long-term use.

Brain-computer interfaces (BCIs) decode neural activity (typically via electroencephalography, EEG) to control external devices such as robotic arms, functional electrical stimulation of paretic muscles, or visual displays, thereby creating a closed-loop neurofeedback system that aims to promote neural reorganization through operant conditioning (McFarland and Wolpaw, 2017; Baniqued et al., 2021). Early-phase clinical trials have demonstrated feasibility and some functional gains, particularly when BCI is combined with robotic assistance or electrical stimulation (Baniqued et al., 2021). However, BCIs face substantial technical challenges: EEG signals are noisy and have low spatial resolution, decoding accuracy is highly variable across individuals and sessions, training users to modulate brain signals is cognitively demanding, and real-time control latencies can be prohibitive. Furthermore, the evidence base remains limited, with small sample sizes, heterogeneous protocols, and unclear mechanisms linking neural decoding to functional recovery (McFarland and Wolpaw, 2017).

Gamified training and virtual reality environments leverage motivational design principles (goal-setting, rewards, progress visualization, narrative engagement) to sustain practice dose and engagement. While these systems can improve adherence and enjoyment relative to conventional exercises, their impact on functional outcomes has been inconsistent (Burdea et al., 2019). A critical limitation is that many gamified systems prioritize surface-level engagement (points, badges, leaderboards) over meaningful motor learning principles, resulting in high-repetition but low-quality practice that does not transfer to real-world dexterity. Moreover, gamification alone does not address the measurement precision problem: even within a game, performance metrics often remain coarse (success/failure, reaction time, trajectory error) rather than revealing fine-grained control dimensions such as force steadiness or timing jitter.

Music- and rhythm-based interventions, including Rhythmic Auditory Stimulation (RAS) and Music-Supported Therapy (MST), have emerged as promising adjuncts to conventional therapy, particularly for gait rehabilitation but increasingly also for upper-limb recovery (Moumdjian et al., 2017; Miendlarzewska and Trost, 2014). The rationale is grounded in neuroscience: auditory rhythms entrain motor timing through strong coupling between auditory perception and motor action mediated by distributed networks including premotor cortex, supplementary motor area, basal ganglia, and cerebellum (Patel, 2008; Thaut et al., 1999). Music also

engages reward and limbic systems, supporting motivation and emotional engagement (Miendlarzewska and Trost, 2014; Altenmüller and Schlaug, 2015). Meta-analyses of RAS for gait have demonstrated moderate to large effect sizes for improving walking speed, stride length, and cadence in stroke survivors and individuals with Parkinson's disease (Moumdjian et al., 2017). For the upper limb, preliminary evidence suggests that MST—structured therapeutic protocols in which patients practice piano playing or percussion with augmented feedback—can improve motor function, timing precision, and patient-reported motivation (Moumdjian et al., 2017; Altenmüller and Schlaug, 2015). However, several critical gaps remain. First, music is often deployed primarily as a *motivational adjunct* rather than as a *quantitative testbed* for probing and measuring specific control dimensions such as millisecond-level timing accuracy, sub-Newton force scaling, or finger individuation. Second, the mechanistic pathways linking musical practice to neuromuscular reorganization and functional recovery remain poorly understood: it is unclear whether benefits arise from the rhythmic structure, the auditory feedback, the motivational appeal, the high practice dose, or some interaction of these factors. Third, music-based interventions have rarely been integrated with objective, high-resolution measurement tools (such as instrumented manipulanda that capture force and timing at high fidelity) or with physiological monitoring (such as high-density EMG) that could reveal how musical training reshapes neuromuscular control strategies.

Movement sonification—the real-time mapping of kinematic variables (position, velocity, acceleration) to auditory parameters (pitch, loudness, timbre)—provides another form of augmented feedback that has shown promise in motor learning and rehabilitation contexts (Moumdjian et al., 2017). By externalizing movement qualities through sound, sonification can enhance error awareness, support attention to movement quality, and provide intuitive, immediate feedback that complements or substitutes for visual information. However, sonification research has primarily focused on reaching and proximal limb control; its application to fine finger dexterity remains underexplored (Moumdjian et al., 2017). Moreover, effective sonification design requires careful consideration of mapping principles, perceptual salience, and cognitive load, and there is limited consensus on which sonification strategies are most effective for which motor learning goals (Moumdjian et al., 2017).

In summary, while technology-assisted rehabilitation has generated considerable enthusiasm and has demonstrated feasibility and preliminary efficacy across multiple modalities, the field remains fragmented, with each approach addressing only a subset of the clinical and scientific challenges. **Three fundamental gaps persist across this landscape**: (1) a lack of *reliable, high-resolution, validated measurement tools* that can quantify timing precision, force scaling, and individuation at the millisecond and sub-Newton scales necessary to reveal subtle deficits and track incremental recovery; (2) a lack of *mechanistic understanding* of the neuromuscular processes (motor unit recruitment, muscle synergies, neural drive dynamics) that underlie behavioral dexterity and how these processes are modified by training or disrupted by stroke; and (3) a lack of *engagement-centered design principles* grounded in users' lived experience, attention, motivation, and sense of agency, which are essential for sustaining high-dose, high-quality practice over the weeks to months required for neuroplastic change. This thesis addresses these three gaps through a tripartite research program—Rhythm Karaoke (RK) for measurement, HD-EMG + AI for mechanism, and ReTouche for engagement—unified by the use of music as a precision-demanding, ecologically valid, and intrinsically motivating testbed.

1.1.4 State of the Art: Measurement, Decoding, and Intervention Technologies for Fine Manual Dexterity

To position the contributions of this thesis within the broader scientific and technological landscape, it is essential to review the current state of the art in three domains directly relevant to the research questions and methods employed here: (1) *objective measurement of manual dexterity*, (2) *neuromuscular decoding and myoelectric control*, and (3) *rhythm- and music-based interventions*. This review is selective rather than exhaustive, focusing on technologies, methods, and findings that directly inform the design, interpretation, and significance of the RK, HD-EMG + AI, and ReTouche projects. *Project-specific implementations are presented in Section 1.3; here we emphasize prior work and gaps.*

Objective Measurement of Manual Dexterity: The Dextrain Manipulandum and Instrumented Assessment

Traditional clinical assessments of upper-limb function—such as the Fugl-Meyer Assessment, Action Research Arm Test, Nine-Hole Peg Test, Box and Block Test, and Jebsen-Taylor Hand Function Test—provide standardized, validated, and widely used metrics for categorizing impairment severity and tracking gross functional recovery (**lyle1981srr**; **platz2005reliability**; **mathiowetz1985adult**; Keh et al., 2022). However, these assessments share several limitations that constrain their utility for research on fine dexterity and precision-based interventions (**wade1983measurement**; **chen2009rasch**). First, they rely on ordinal scoring systems or count-based metrics (e.g., number of pegs placed, time to complete a task) that lack the resolution to detect small but functionally meaningful differences in control quality (**chen2009rasch**). Second, they typically assess task completion or outcome (success/failure, time) rather than the underlying control processes (how force is modulated, how timing is coordinated, how individuation is achieved). Third, they are administered infrequently (often only at baseline and endpoint in clinical trials) and thus cannot capture session-to-session variability or fine-grained recovery trajectories (**wade1983measurement**).

In response to these limitations, researchers have developed *instrumented assessment tools* that use sensors, robotics, or manipulanda to capture kinematic, kinetic, and temporal variables at high resolution during standardized motor tasks. One exemplar of this approach is the **Dextrain Finger Force Manipulandum**, a patented device designed specifically to quantify multi-digit timing and force coordination with high fidelity (Térémetz et al., 2015; Lindberg, Térémetz, and Maier, 2016). The Dextrain system comprises five independent, spring-loaded force sensors positioned to accommodate the distal pads of all five digits. Each sensor measures applied force with 0.01 Newton resolution and records force-time profiles at sampling rates exceeding 100 Hz, enabling precise capture of force onset timing, peak force amplitude, force steadiness (quantified as the coefficient of variation or standard deviation during sustained contractions), and inter-digit coordination. Validation studies have demonstrated that Dextrain-derived metrics correlate with clinical measures of hand function and are sensitive to differences between healthy controls, neurological patients, and individuals with aging-related dexterity decline (Térémetz et al., 2015). Critically, the Dextrain provides *continuous, quantitative* data rather than binary pass/fail judgments, enabling detailed analyses of how individuals scale forces across different target levels, how precisely they time force onsets relative to external cues, and how independently they control individual fingers.

Beyond the Dextrain, other instrumented dexterity assessment systems have been developed, including robotic platforms such as the KINARM, the MIT-MANUS, and haptic interfaces that measure reaching kinematics, grip force, and fingertip contact dynamics (**krebs1998rehabilitation**; **scott1999apparatus**; **dukelow2010quantitative**). These systems have contributed valuable insights into post-stroke motor control, revealing deficits in trajectory smoothness, force-rate coupling, and grip-load force coordination (**lang2005quantification**; Keh et al., 2022). However, most of these systems focus on reaching and proximal limb control or on gross grip force rather than on the fine-grained, multi-digit timing and individuation that are central to skilled manual dexterity (**raghavan2010compensatory**). Moreover, many instrumented platforms are large, expensive, and confined to laboratory or clinical settings, limiting their accessibility for widespread use.

The state of the art in objective dexterity measurement thus provides a foundation: the Dextrain Manipulandum offers validated, high-resolution quantification of force and timing, but applications have been limited primarily to static force tasks or simple rhythmic tapping. Building on this, the present thesis explores integrating Dextrain with music-based rhythmic synchronization tasks and systematic auditory feedback manipulations to create a dynamic, ecologically valid, and experimentally controllable platform for probing sensorimotor synchronization, error correction, and adaptive timing (see Section 1.3.1).

Neuromuscular Decoding and Myoelectric Control: From Surface EMG to High-Density Arrays and AI-Driven Regression

Electromyography (EMG)—the recording of electrical activity generated by muscle fibers during contraction—has been used for decades in clinical neurology, kinesiology, and rehabilitation research to assess muscle function, diagnose neuromuscular disorders, and study motor control. Conventional surface EMG (sEMG) employs a small number of electrodes (typically 1-8 channels) placed over target muscles to record the aggregate electrical activity of motor units within the pickup volume. While sEMG provides valuable qualitative information about muscle activation timing and relative intensity, it suffers from limited spatial resolution, crosstalk from adjacent muscles, and difficulty in resolving individual motor unit activity. These limitations have constrained the utility of conventional sEMG for fine-grained analysis of muscle coordination and for real-time myoelectric control applications (Kamper et al., 2006; De Luca et al., 2009).

Over the past two decades, advances in electrode technology, multichannel amplifiers, and signal processing have enabled the development of **high-density surface EMG (HD-EMG)** systems that employ arrays of tens to hundreds of small electrodes arranged in grids or semi-circular configurations around muscles or muscle groups (Kamper et al., 2006; De Luca et al., 2009). HD-EMG offers several advantages over conventional sEMG: (1) higher spatial resolution, enabling localization of motor unit activity and decomposition of the interference EMG signal into individual motor unit action potential trains; (2) the ability to map spatial activation patterns and identify muscle synergies (coordinated recruitment of multiple muscles); (3) improved signal-to-noise ratio through spatial filtering and common-mode rejection; and (4) the capacity to extract rich feature sets for machine learning-based decoding of movement intent and force output. HD-EMG has been successfully applied to decode hand and finger movements, with studies demonstrating that

HD-EMG signals can be used to classify discrete gestures, predict continuous finger kinematics, and estimate multi-degree-of-freedom (multi-DOF) forces (Kamper et al., 2006; De Luca et al., 2009).

A particularly relevant line of research for this thesis concerns **AI-driven regression models for continuous force decoding** from EMG. Early myoelectric control systems, developed primarily for prosthetic applications, relied on pattern recognition (classification) approaches that mapped EMG features to discrete movement classes (e.g., hand open, hand close, pinch grip). While classification is sufficient for prosthetic control with a limited set of pre-defined grasps, it is inadequate for natural, dexterous control that requires simultaneous, proportional modulation of multiple DOFs (e.g., independent force control of each finger). To address this, researchers have developed *regression-based decoding methods* that predict continuous kinematic or kinetic outputs (finger positions, joint angles, fingertip forces) from EMG signals (Kamper et al., 2006; De Luca et al., 2009). Linear regression models (including ridge regression and partial least squares) provide interpretable mappings from EMG features to force outputs and can achieve moderate decoding accuracy with low computational cost (Kamper et al., 2006). However, the relationship between EMG amplitude and force is inherently nonlinear (due to motor unit recruitment, rate coding, and muscle mechanics), and linear models often underperform when generalizing across force levels, contraction types, or individuals.

To overcome these limitations, researchers have increasingly turned to *nonlinear machine learning models*, including random forests, support vector machines, and deep neural networks. Convolutional neural networks (CNNs) can learn spatial features from multi-channel EMG arrays, while recurrent neural networks (RNNs), long short-term memory (LSTM) networks, and more recently, transformer architectures can capture temporal dependencies and dynamics in EMG-force relationships (Montazerin et al., 2023; Montazerin, 2023). These deep learning approaches have demonstrated superior decoding performance (e.g., higher coefficient of determination R^2, lower root-mean-square error) compared to classical methods, particularly for complex, multi-DOF tasks (Montazerin et al., 2023). However, deep models require large training datasets, careful tuning of hyperparameters, and substantial computational resources, and they often function as "black boxes" that provide little mechanistic insight into which muscles or features drive predictions.

A critical challenge for all EMG decoding methods—whether classification or regression, linear or nonlinear—is *generalization*: the ability of models trained on one set of conditions or individuals to perform accurately on new conditions, tasks, or users. Most EMG decoding studies have focused on within-session or within-participant performance, where models are trained and tested on data from the same individual, often within the same recording session. However, EMG signals exhibit substantial variability across sessions (due to electrode placement shifts, skin impedance changes, fatigue) and across individuals (due to anatomical differences, muscle strength, motor strategies). Developing models that generalize *across users without requiring per-user calibration* is essential for clinical scalability: in time-constrained rehabilitation settings, it is impractical to collect extensive training data and retrain models for each patient. Recent work has begun to address this challenge through transfer learning, domain adaptation, and the use of spatial or temporal normalization techniques, but robust cross-user generalization remains an open problem (Montazerin et al., 2023; Montazerin, 2023).

Another underexplored aspect of HD-EMG for hand control is the inclusion of *intrinsic hand muscles* (thenar, hypothenar, interossei, lumbricals) in the recording montage. Most prior studies have focused on forearm muscles (flexor digitorum

superficialis, extensor digitorum, etc.) because they are larger, more accessible, and generate higher-amplitude EMG signals. However, intrinsic hand muscles play a critical role in fine finger individuation and force modulation, and their activity is often altered after stroke (Kamper et al., 2006; Keh et al., 2022). **A distinctive feature of the HD-EMG + AI project in this thesis is the use of a 224-electrode array that covers both forearm and intrinsic hand muscles**, providing unprecedented spatial resolution and the potential to decode fine, individuated finger forces with higher fidelity than previous systems.

Finally, for EMG decoding to be useful in real-time biofeedback or rehabilitation applications, it must operate with *low latency* (ideally <50 milliseconds end-to-end) to avoid disrupting the user's sense of agency and to enable closed-loop motor learning. Achieving real-time performance requires optimized signal acquisition, efficient feature extraction, fast inference (particularly challenging for deep neural networks), and careful management of computational resources. Recent work targets proof-of-concept decoding with $R^2 > 0.75$ and latencies <50 ms, positioning such systems as suitable for real-time biofeedback applications (e.g., EMG-driven visual displays or EMG-to-music interfaces). In this thesis, we investigate this approach in a high-density setting (see Section 1.3.2).

In summary, the state of the art in neuromuscular decoding provides powerful tools for linking muscle activity to motor output, but significant gaps remain: most systems do not include intrinsic hand muscles, few achieve robust cross-user generalization, and real-time, low-latency operation remains challenging. In this thesis, we pursue an integration of high-density recording (including intrinsic muscles), AI-driven regression for continuous force prediction, and real-time operation within a music-based rehabilitation context that leverages the Dextrain Manipulandum as the ground-truth force reference (see Section 1.3.2).

Rhythm- and Music-Based Interventions: Evidence, Mechanisms, and Gaps

The therapeutic use of music and rhythm in neurorehabilitation has a long history, rooted in observations that musical activities can engage, motivate, and structure motor behavior in ways that conventional exercises do not. Over the past two decades, this empirical tradition has been increasingly informed by neuroscience research demonstrating strong coupling between auditory perception and motor action, particularly for rhythmic and temporal aspects of movement (Patel, 2008; Thaut et al., 1999; Miendlarzewska and Trost, 2014). Several specific intervention modalities have emerged, each with distinct protocols, mechanisms, and evidence bases.

Rhythmic Auditory Stimulation (RAS) is a neurologic music therapy technique in which external auditory rhythms (typically metronome beats or rhythmic music) are used to cue and entrain motor movements, most commonly in gait rehabilitation for stroke, Parkinson's disease, and traumatic brain injury (Thaut et al., 1999; Moumdjian et al., 2017). The core principle is that auditory rhythms provide a stable temporal template that can scaffold motor timing, reduce variability, and facilitate the entrainment of motor oscillators in the central nervous system (involving basal ganglia, cerebellum, and supplementary motor area).

Meta-analyses of RAS for gait have consistently shown moderate to large effect sizes: stroke survivors who practice walking with RAS exhibit significant improvements in walking speed (cadence), stride length, gait symmetry, and temporal regularity compared to controls who practice without auditory cues (Moumdjian et al., 2017). The mechanisms underlying these benefits likely include: (1) direct entrainment of spinal central pattern generators and cortical motor regions by auditory

rhythms, (2) increased saliency and predictability of movement timing cues, (3) offloading of internal timing processes (which may be impaired after stroke) onto external, reliable cues, and (4) enhanced attention and motivation sustained by rhythmic auditory structure (Patel, 2008; Thaut et al., 1999).

While RAS has been primarily applied to gait, there is growing interest in extending rhythmic cueing to upper-limb movements. Preliminary studies have shown that rhythmic auditory cues can improve the timing, smoothness, and amplitude of reaching movements and can facilitate bimanual coordination in stroke survivors (Moumdjian et al., 2017). However, the evidence base for upper-limb RAS is less robust than for gait, and the specific parameters (tempo, rhythmic complexity, feedback modality) that optimize upper-limb training remain unclear.

Music-Supported Therapy (MST) and **Music-Based Motor Rehabilitation** represent broader therapeutic frameworks in which patients engage in structured musical activities—such as playing piano, percussion instruments, or electronic drum pads—as a medium for motor training (**schneider2007music**; **altenmullergruhn2015music**; Altenmüller and Schlaug, 2015; Miendlarzewska and Trost, 2014). The rationale for MST extends beyond rhythmic entrainment to encompass multiple mechanisms: (1) music enhances both performance and engagement through neurobiological and motivational pathways: empirical data from the Rhythm Karaoke project demonstrate that melodic content significantly improves timing precision (up to 42% error reduction for familiar melodies such as "Balou", $t(29) = 13.17$, $p < 0.001$) and that speech-like stimuli are rated as less difficult and more satisfying than abstract tones on Intrinsic Motivation Inventory scales, despite equivalent performance demands (Arslan, Xiao, and Lindberg, 2025); these effects likely reflect engagement of dopaminergic reward circuits and limbic regions that support sustained practice and positive affect (**salimpoor2011anatomically**; Miendlarzewska and Trost, 2014); (2) musical tasks demand fine-grained timing, force modulation, and finger individuation, providing ecologically valid, goal-directed practice of dexterous control (**schneider2007music**); (3) music is structured hierarchically (meter, phrase structure), which may support chunking, motor sequence learning, and anticipatory planning (**koelsch2012neural**); and (4) musical feedback provides immediate, intuitive knowledge of performance (auditory results reflect movement quality), supporting error detection and correction (Altenmüller and Schlaug, 2015; Miendlarzewska and Trost, 2014; Patel, 2008).

Several pilot studies and small-scale randomized controlled trials have investigated MST for upper-limb rehabilitation after stroke. For example, studies have shown that stroke survivors who practice piano with or without augmented visual feedback exhibit improvements in finger tapping speed, dexterity scores (e.g., Nine-Hole Peg Test), and upper-limb function scales (e.g., Fugl-Meyer) compared to controls who receive conventional therapy (**schneider2007music**; **ripollehernandez2016music**; Altenmüller and Schlaug, 2015; Miendlarzewska and Trost, 2014).

Neuroimaging studies (fMRI, EEG) have demonstrated that musical training is associated with increased activation in motor cortex, premotor cortex, and sensorimotor integration regions, as well as enhanced structural and functional connectivity in cortico-cerebellar and cortico-striatal networks (Patel, 2008; Miendlarzewska and Trost, 2014). However, the evidence base for MST remains limited by small sample sizes, heterogeneous protocols, lack of standardized outcome measures, and insufficient understanding of dose-response relationships and the relative contributions of rhythmic structure, auditory feedback, motivational engagement, and motor practice dose to observed benefits.

A critical gap in the music-based intervention literature is that **music is typically used for its motivational or affective benefits rather than as a precision-based testbed for quantifying and targeting specific control dimensions**. Most studies assess outcomes using clinical scales or gross motor metrics (e.g., number of notes played, time to complete a piece) rather than fine-grained measures of timing accuracy (asynchrony, jitter, phase correction), force scaling (smoothness, steadiness, proportionality), or individuation (co-contraction, motor overflow). As a result, it remains unclear whether musical training improves precision in these specific domains and whether such improvements generalize to non-musical functional tasks. Furthermore, the field lacks validated, reliable, music-based assessment tools that could serve as outcome measures in clinical trials or as diagnostic instruments for quantifying dexterity deficits.

Movement sonification and **augmented auditory feedback** represent another avenue through which auditory information is used to enhance motor learning and rehabilitation. Sonification maps kinematic variables (e.g., hand velocity, trajectory error, smoothness) to auditory parameters (pitch, loudness, timbre), providing real-time, intuitive feedback about movement quality (Moumdjian et al., 2017). Studies have shown that sonification can improve reaching accuracy, movement smoothness, and bimanual coordination in healthy learners and neurological populations (Moumdjian et al., 2017). However, sonification research has primarily focused on reaching and proximal limb control; applications to fine finger dexterity, timing precision, and force modulation are underexplored. Moreover, effective sonification design requires careful consideration of perceptual mappings, cognitive load, and user preferences, and there is limited consensus on optimal design principles.

In summary, rhythm- and music-based interventions have demonstrated promise for motor rehabilitation, supported by converging neuroscience evidence for auditory-motor coupling and motivational engagement. However, critical gaps remain: (1) music is rarely used as a *quantitative testbed* with millisecond-level timing metrics and controlled feedback manipulations; (2) the mechanisms linking musical practice to neuromuscular reorganization are poorly understood; and (3) there is a lack of validated, music-based measurement tools suitable for clinical trials and personalized rehabilitation. This thesis addresses these gaps by using music as a precision-demanding assessment context with controlled feedback and by validating timing metrics suitable for clinical research (see Section 1.3.1).

Synthesis: Technology Landscape and Positioning of This Thesis

The preceding review reveals that the fields of dexterity measurement, neuromuscular decoding, and music-based rehabilitation have each made substantial progress, yet they remain largely disconnected, and each faces persistent limitations. These advances and limitations are synthesized and critically analyzed in Tables 1.1 (Measurement and Decoding) and 1.2 (Training and Feedback), presented below. These tables provide a structured overview of representative technologies spanning conventional therapy, objective dexterity measurement, HD-EMG decoding, rhythm/music-based interventions, movement sonification, and brain-computer interfaces, highlighting the domains addressed, sensors employed, feedback modalities, underlying principles, key references, strengths, and limitations of each approach.

Collectively, this landscape reveals **three fundamental gaps** that motivate the integrated, three-lens architecture of this thesis:

Gap 1: Lack of Reliable, High-Resolution, Music-Based Timing Precision Measures. While the Dextrain Manipulandum provides validated force and timing quantification, and music-based interventions show promise for motor rehabilitation, there exists no validated, psychometrically sound, music-based assessment that measures timing precision at millisecond resolution, is sensitive to auditory feedback manipulations, and has established test-retest reliability and convergent validity with clinical dexterity measures. **The RK project fills this gap** by developing a music-based rhythmic synchronization protocol on the Dextrain, extracting detailed timing metrics (asynchrony, variability, error correction, phase drift), validating these metrics psychometrically (ICC, confidence intervals, minimal detectable change), and demonstrating sensitivity to experimental manipulations and clinical impairment.

Gap 2: Lack of Mechanistic Understanding Linking Behavioral Precision to Neuromuscular Control. Behavioral metrics, however precise, remain descriptive: they reveal *what* is impaired but not *why* or *how* at the neuromuscular level. HD-EMG offers unprecedented spatial resolution of muscle activation, and AI-driven regression enables continuous force decoding, but prior work has not integrated these capabilities with fine-grained behavioral measurement of timing and force in music-based tasks, nor has it included intrinsic hand muscles or achieved robust cross-user generalization and real-time operation. **The HD-EMG + AI project fills this gap** by recording 224-channel HD-EMG (including intrinsic hand muscles) synchronously with Dextrain force measurements during the RK protocol, developing AI regression models that decode continuous finger forces with high accuracy ($R^2 > 0.75$) and low latency (<50 ms), and linking neuromuscular activation patterns (co-contraction, synergies, motor unit dynamics) to behavioral timing precision. This provides both mechanistic insight (explaining precision deficits) and a pathway to clinical translation (EMG-driven biofeedback, exploratory EMG-to-music interfaces).

Gap 3: Lack of Engagement-Centered Design Principles Grounded in Lived Experience. Even the most sophisticated measurement and mechanistic models will have limited clinical impact if individuals do not engage with training tasks over sufficient durations and with sufficient quality of practice. Yet rehabilitation technology research has historically neglected the subjective, phenomenological dimensions of practice: how feedback shapes attention, how errors are experienced and corrected, how motivation and agency emerge and are sustained. **The ReTouche project fills this gap** by conducting a phenomenological and HCI inquiry into the lived experience of learning fine finger coordination in a musical context, using autoethnography, focus groups, and interviews to generate rich qualitative insights. Thematic analysis reveals that embodied visual feedback externalizes errors, feedback reorients attention from outcomes to processes, users develop personalized strategies and metacognitive awareness, and sustained engagement is driven by visible progress and agency rather than gamification. These findings provide evidence-based design principles for interfaces that foster long-term engagement, directly applicable to rehabilitation technology.

By addressing these three gaps in an integrated manner—measurement (RK), mechanism (HD-EMG + AI), and engagement (ReTouche)—this thesis advances toward a coherent, precision-based, user-centered framework for rehabilitation of fine manual dexterity after stroke. The following section ("Why Music as a Precision Testbed?") will elaborate on the specific rationale for using music as the unifying context for this research program, grounding this choice in the neuroscience of auditory-motor coupling, the structured temporal demands of musical tasks, and the intrinsic motivational appeal of musical engagement.

Table 1.1: Measurement and Decoding Technologies — State of the Art and Limitations

The technologies reviewed in Section 1.4 for measuring manual dexterity and decoding neuromuscular control are synthesized in Table 1.1 below. This table highlights the targeted capabilities, metrics/feedback modalities, quantitative resolution (key figures), recognized strengths, persistent limitations (gaps), level of evidence, and key supporting references for each approach. The limitations column reveals the three fundamental gaps that motivate this thesis: **[Gap 1: RK]** lack of millisecond-level timing precision in clinical assessments, **[Gap 2: HD-EMG]** absence of continuous force decoding with intrinsic hand muscle inclusion in HD-EMG systems, and insufficient integration of high-resolution measurement with mechanistic neuromuscular modeling.

TABLE 1.1: Measurement and Decoding: Strengths and Gaps. UL = upper limb; HD-EMG = high-density EMG; DOF = degrees of freedom; FMA = Fugl-Meyer Assessment; LoE = Level of Evidence.

Approach (Domain)	**Targeted Capability**	**Metrics/ Feedback**	**Key Figures / Resolution**	**Strengths (Advantages)**	**Limitations (Gaps)**	**LoE**	**Key References**
Clinical standard assessments (UL/Hand)	Global UL function, Force	FMA, Box & Block Test, Nine-Hole Peg Test	Ordinal scale; Low resolution ($\approx$ 10 pts) (Voet et al., 2019)	Widely validated; Established norms; Assesses general severity (Keh et al., 2022)	Limited resolution (no ms timing, no sub-Newton force); Insensitive to fine deficits (timing, force scaling, individuation) (De Luca et al., 2009; Parker, Wade, and Langton Hewer, 1986). **[Gap 1: RK]**	Validation/ Norms	(Voet et al., 2019; Keh et al., 2022; Térémetz et al., 2015)
Dextrain Manipulandum (Platform)	Fine multi-dimensional dexterity (Timing, Force, Independence)	Force (N); Timing (ms); Jitter, Autocorrelation	Force: **0.01 N**; Timing: **1 kHz**; Multi-digital (Térémetz et al., 2015)	Precise quantification; High temporal and kinetic resolution; Validated for stroke/healthy discrimination (Térémetz et al., 2015)	Non-portable; Primarily static tasks; Not yet integrated with rhythmic music tasks (RK) or EMG decoding (Térémetz et al., 2015). **[Gap 1: RK]**	Validation Study	(Térémetz et al., 2015)
HD-EMG (General Literature)	Neural Drive, Gesture classification	Spatial muscle activation (EMG); Accuracy	Typically **64–128 ch**; Classification accuracy $\approx$ **90%** (Malešević et al., 2021; Montazerin et al., 2023)	High spatial resolution; Reveals synergies (Farina et al., 2010)	Often limited to classification (not continuous/proportional); Excludes most intrinsic hand muscles (Malešević et al., 2021; Tacca et al., 2024a). **[Gap 2: HD-EMG]**	Review/ Database	(Farina et al., 2010; Malešević et al., 2021; Tacca et al., 2024b)
AI for Force Estimation (Literature)	Continuous force/kinematics prediction	Regression (SVR, CNN, LSTM); R^2	R^2 varies ($\approx$ 0.7–0.9); Latency often **>100 ms** (Li, Shi, and Yu, 2021)	Proportional control possible (Muceli and Farina, 2012)	Latency too high for real-time biofeedback (<50 ms) (Englehart and Hudgins, 2003); Generalization difficult (Montazerin, 2023). **[Gap 2: HD-EMG]**	Protocol/ Pilot	(Muceli and Farina, 2012; Li, Shi, and Yu, 2021; Englehart and Hudgins, 2003)

Approach (Domain)	Targeted Capability	Metrics/ Feedback	Key Figures / Resolution	Strengths (Advantages)	Limitations (Gaps)	LoE	Key References
HD-EMG + AI (Thesis Project)	Continuous proportional decoding of finger forces	IA Regression (CNN-LSTM); Biofeedback Force/Timing	**224 electrodes** (incl. **64 intrinsic**) (confirmed, 2024); Target: $R^2 > 0.75$; Latency target **<50 ms** (confirmed, 2024)	Intrinsic muscle coverage; Continuous force prediction; Real-time operation for neuromuscular feedback (confirmed, 2024)	Not yet clinically deployed; Requires interface (ReTouche) to ensure sustained engagement (confirmed, 2024). **[Gap 3: ReTouche]**	Protocol/ Manuscrip	(confirmed, 2024; Montazerin et al., 2023)

Table 1.2: Training and Feedback Interventions — State of the Art and Limitations

The intervention and feedback technologies reviewed in Section 1.4 are synthesized in Table 1.2 below, which highlights targeted capabilities, metrics/feedback modalities, quantitative key figures (dose, effect sizes), recognized strengths, persistent limitations (gaps), level of evidence, and key references. The limitations column reveals critical gaps in dose/adherence sustainability **[Gap 3: ReTouche]**, precision of timing metrics in music-based interventions **[Gap 1: RK]**, and integration of mechanistic neuromuscular feedback **[Gap 2: HD-EMG]** informed by users' lived experience.

TABLE 1.2: Training and Feedback: Strengths and Gaps. UL = upper limb; RAS = rhythmic auditory stimulation; MST = music-supported therapy; KR/KP = knowledge of results/performance; RCT = randomized controlled trial; SR = systematic review; LoE = Level of Evidence.

Approach (Domain)	**Targeted Capability**	**Metrics/ Feedback**	**Key Figures / Resolution**	**Strengths (Advantages)**	**Limitations (Gaps)**	**LoE**	**Key References**
Conventional Therapy (High Dose)	Motor Recovery, Dose	Intense repetitions; Functional scales	High dose (e.g., **300+ reps/day**) (Lee et al., 2021)	Established efficacy for global function (dose-dependent) (Lee et al., 2021)	Low and unsustainable adherence (often **<30%** at home) (Weegen et al., 2016; Verrienti et al., 2023). **[Gap 3: ReTouche]**	SR/Meta	(Schmidt and Lee, 2011; Lee et al., 2021)
RAS/MST (Rhythmic/Music)	Coordination, Motivation, UL Function	Auditory Rhythmic Cueing; FMA, 9-HPT	Gait speed improvement Δ: **0.14 ± 0.03 m/s** (Awad et al., 2024)	Established efficacy for gait; Improves UL function and motivation (Moumdjian et al., 2017; Raghavan et al., 2016; Amengual et al., 2013)	Lacks precise timing and force measurement; Weak LoE for fine dexterity (Huang et al., 2021). **[Gap 1: RK]**	Meta/SR	(Awad et al., 2024; Moumdjian et al., 2017; Schauer and Mauritz, 2003)
Movement Sonification (Arm)	Movement Quality, Error Correction	Auditory feedback KP/KR (pitch, tone)	Reduces abnormal movements (e.g., **34% reduction** in pilot $N = 20$) (Douglass-Kirk et al., 2023; Peyre et al., 2023)	Reduces compensations; Promotes attention to movement quality (Chen, Fujii, and Schlaug, 2016)	Primarily proximal movements (arm); Requires neuromuscular metrics (HD-EMG) for targeted feedback. **[Gap 2: HD-EMG]**	Pilot/ Case Series	(Peyre et al., 2023; Douglass-Kirk et al., 2023)
Game/Robotic Interfaces (e.g., MusicGlove)	Repetition, Dose, Motivation	Haptic gloves/sensors; Visual/auditory feedback	High dose (e.g., **466 grips/day** for MusicGlove) (Zondervan et al., 2016)	Increases dose; Strong initial motivation (Urbina, Abarca, and Elias, 2024)	Focuses on quantity; Lacks sustained engagement (novelty effect); Design often cognitivist (vs. embodied) (Zondervan et al., 2016). **[Gap 3: ReTouche]**	RCT/Pilot	(Friedman et al., 2014; Zondervan et al., 2016)

Approach (Domain)	Targeted Capability	Metrics/ Feedback	Key Figures / Resolution	Strengths (Advantages)	Limitations (Gaps)	LoE	Key References
ReTouche (Thesis Project)	Adherence, Strategy, Agency, User Experience (UX)	Autoethnography, CSO, Focus Group; Embodied representations	CSO $N = 18$, Autoethnography $N = 3$ (2 weeks), Focus Group $N = 4$ (Author(s), 2026)	Reveals how design shapes attention and strategy; Sustained engagement via agency.	Qualitative evaluation; Requires integration with quantitative metrics (RK/HD-EMG) to validate clinical gains. **[Gap 1/2]**	Qual. HCI	(Author(s), 2026; Kirsh, 2013; Mackay and McGrenere, 2025)

1.2 Why Music as a Precision Testbed?

1.2.1 Why Music as a Precision Testbed?

The preceding review has identified three fundamental gaps in the current landscape of rehabilitation technology and research: (1) lack of reliable, high-resolution measurement tools that quantify timing precision at millisecond resolution; (2) lack of mechanistic understanding linking observable behavioral deficits to underlying neuromuscular control processes; and (3) lack of engagement-centered design principles grounded in the lived experience of users. To address these gaps, this thesis adopts a unified methodological and conceptual framework centered on *music*—not as a mere motivational adjunct or entertainment overlay for conventional rehabilitation exercises, but as a *precision-demanding testbed* in which perception-action coupling can be rigorously probed, perturbed, and measured with experimental precision comparable to laboratory psychophysics, while retaining ecological validity and intrinsic motivational appeal.

The choice of music as the unifying context for this research program is deliberate and grounded in converging evidence from neuroscience, motor control, music cognition, and rehabilitation science. Music provides a rare combination of attributes that are essential for advancing both the scientific understanding and clinical management of fine manual dexterity after stroke: (1) music imposes *structured temporal demands* that require millisecond-level timing precision and afford experimental manipulation of rhythmic cues and feedback; (2) music engages powerful *auditory-motor coupling* mechanisms that entrain motor timing, facilitate error correction, and recruit distributed brain networks supporting sensorimotor integration; (3) music is *ecologically valid and intrinsically rewarding*, addressing the persistent challenge of sustaining engagement and adherence over extended training periods; and (4) music provides a naturalistic context in which timing, force, and individuation are all task-relevant and measurable simultaneously, enabling holistic study of their interplay.

Critically, we position music not as a motivational tool alone but as a *quantitative experimental paradigm*—a domain in which the precision demands of musical performance can be harnessed to reveal subtle deficits, track recovery trajectories, and investigate the neuromuscular mechanisms underlying dexterous control with rigor comparable to controlled laboratory tasks. This section elaborates on the scientific rationale for this stance in three parts: (1) the structured temporal demands of music and their implications for sensorimotor synchronization research, (2) the neuroscience evidence base for music-based interventions and the critical gap that music-as-measurement fills, and (3) the role of auditory feedback as an experimental lever for probing internal control processes and informing rehabilitation design.

Structured Temporal Demands: Music as a Scaffold for Sensorimotor Synchronization

Music is fundamentally a temporal art form: its structure is defined not merely by the pitch content of notes but, more fundamentally, by the temporal relationships among events—the durations of notes, the intervals between onsets, the organization of beats into meter, and the hierarchical nesting of rhythmic patterns across multiple timescales (Patel, 2008; Palmer and Krumhansl, 1990). This temporal structure imposes precise coordination demands on performers, who must time their movements (finger presses, bows, breath onsets) to align with intended note onsets, maintain steady tempos or intentionally modulate timing for expressive purposes, and

synchronize with co-performers or external rhythmic cues (Repp, 2005; Repp and Su, 2013). These demands are not vague or subjective but are quantifiable at millisecond resolution: skilled musicians routinely achieve mean asynchronies (temporal offsets between intended and actual note onsets) of less than 30 milliseconds when synchronizing to a metronome, and they detect and correct timing errors on the order of 10-20 milliseconds (Repp, 2005; Repp and Su, 2013).

The hierarchical temporal structure of music—organized into beats (typically 400-800 ms inter-onset intervals), measures (groupings of beats defined by meter), and phrase-level structures—provides a rich scaffold for investigating sensorimotor synchronization (SMS) across multiple timescales (Palmer and Krumhansl, 1990; Patel, 2008). SMS refers to the rhythmic coordination of movement with an external temporal signal, most commonly studied in finger-tapping paradigms where participants tap in synchrony with an isochronous metronome or rhythmic sequence (Repp and Su, 2013; Repp, 2005). SMS tasks have been extensively used in cognitive neuroscience and motor control research because they provide a controlled, quantifiable context for studying the perception-action coupling, internal timekeeping, error correction, and adaptive synchronization processes that are fundamental to coordinated behavior (Repp and Su, 2013). Music extends beyond simple isochronous tapping by introducing hierarchical accent structures (strong beats, downbeats, phrase boundaries), rhythmic complexity (syncopation, polyrhythm), tempo modulations, and melodic-harmonic content that engage higher-level cognitive and motor processes including anticipation, chunking, sequence planning, and auditory-motor integration at multiple representational levels (Palmer and Krumhansl, 1990; Patel, 2008).

The experimental advantages of music-based SMS paradigms are manifold. First, the structured temporal demands of music make *timing precision* a salient and task-relevant variable that can be measured objectively and continuously throughout performance. Unlike many functional tasks (e.g., reaching for a cup, buttoning a shirt) where timing constraints are soft or variable, musical performance demands tight temporal coupling to external cues or internal rhythmic templates, and deviations from target timing are immediately detectable both by the performer (through auditory feedback) and by objective measurement systems. This makes music an ideal context for revealing subtle timing deficits that might be masked or compensated for in less temporally constrained tasks (Repp and Su, 2013).

Second, the hierarchical rhythmic structure of music affords *controlled experimental manipulations* that can systematically probe sensorimotor integration and adaptive control. For instance, researchers can manipulate inter-onset intervals (speeding or slowing tempo), introduce tempo perturbations (sudden phase shifts, tempo ramps), alter accent patterns (emphasizing different beats to create different metrical interpretations), delay auditory feedback (creating a temporal mismatch between action and sensory consequence), or remove auditory feedback entirely (requiring reliance on internal timekeeping and proprioceptive/tactile cues) (Repp and Su, 2013; Repp, 2005). Each of these manipulations provides insight into different aspects of the sensorimotor control architecture: tempo perturbations reveal phase-correction mechanisms (how rapidly and effectively the motor system adjusts to timing errors); feedback delays reveal the reliance on auditory reafference for online error correction; and removal of auditory cues reveals the stability and accuracy of internal timekeeping processes (Repp and Su, 2013). Such manipulations are difficult to implement in less structured tasks and would lack the intuitive interpretability that musical context provides.

Third, musical tasks naturally integrate *multiple control dimensions*—timing, force,

and individuation—within a single, ecologically valid activity. Piano performance, for example, requires not only precise timing of key presses but also graded modulation of keystroke force (velocity) to produce dynamics ranging from pianissimo to fortissimo, as well as independent control of individual fingers to play chords, melodies, and polyphonic textures (Patel, 2008). This multidimensional demand profile mirrors the complexity of many real-world manual tasks and allows researchers to study the interplay and trade-offs among these control dimensions (e.g., how does increasing tempo affect force steadiness? how does the need for individuation interact with timing precision?) in a naturalistic context rather than through isolated, artificial laboratory tasks that assess each dimension in isolation (Palmer and Krumhansl, 1990).

Neuroscience research has provided converging evidence that musical training and performance engage and modulate the neural circuits underlying sensorimotor synchronization, motor timing, and fine motor control. Neuroimaging studies using functional magnetic resonance imaging (fMRI), electroencephalography (EEG), and magnetoencephalography (MEG) have consistently demonstrated that musical activities—particularly rhythmic synchronization and performance—activate a distributed network of brain regions including the primary motor cortex (M1), premotor cortex (PMC), supplementary motor area (SMA), basal ganglia (particularly the striatum and putamen), cerebellum, and auditory cortex (Zatorre, Chen, and Penhune, 2007; Patel, 2008; Thaut et al., 1999). Critically, these regions are not merely co-activated during music performance but exhibit *functional coupling*: their activity is temporally correlated, suggesting that they form an integrated network supporting auditory-motor transformation, temporal prediction, and error correction (Zatorre, Chen, and Penhune, 2007; Patel, 2008).

The basal ganglia and cerebellum, in particular, are thought to play complementary roles in motor timing: the basal ganglia contribute to beat-based, internally generated timing (the "clock" that supports regular, rhythmic movements even in the absence of external cues), while the cerebellum supports event-based timing and adaptive error correction (rapidly adjusting movement timing in response to discrepancies between predicted and actual sensory feedback) (Patel, 2008; Thaut et al., 1999). Musical training has been shown to enhance the structural and functional properties of these timing networks. For example, professional musicians exhibit increased gray matter volume in motor and auditory cortices, enhanced white matter connectivity in cortico-cerebellar and cortico-striatal pathways, and increased functional connectivity between auditory and motor regions during rest and task performance (Zatorre, Chen, and Penhune, 2007; Miendlarzewska and Trost, 2014). These neuroplastic changes are associated with behavioral advantages in timing precision, sensorimotor synchronization, and fine motor control that extend beyond musical tasks to non-musical perceptual and motor domains (Miendlarzewska and Trost, 2014).

Importantly, musical training effects are not limited to professional musicians or lifelong practitioners. Short-term interventions (weeks to months of structured musical practice) in non-musicians have been shown to induce measurable neuroplastic changes and behavioral improvements in timing, rhythm discrimination, and motor coordination (Miendlarzewska and Trost, 2014). These findings suggest that music-based training protocols could be leveraged in rehabilitation contexts to drive adaptive reorganization of sensorimotor networks and improve timing precision even in individuals with neurological impairments such as stroke, provided that training is structured, dose-appropriate, and targets specific control dimensions with high-resolution feedback.

For the purposes of this thesis, the structured temporal demands of music provide the foundation for the **Rhythm Karaoke (RK) project**, which uses music-based rhythmic synchronization tasks on the Dextrain Finger Force Manipulandum to quantify timing precision at millisecond resolution. The RK protocol leverages the hierarchical rhythmic structure of music and speech-like stimuli to impose tight timing constraints, systematically manipulates auditory feedback (synchronous, delayed, altered accents, and silence) to probe error-correction dynamics and the reliance on external vs. internal timing cues, and extracts a rich set of timing metrics (mean asynchrony, variability, lag-1 autocorrelation, phase drift) that are psychometrically validated through test-retest reliability analyses (Repp and Su, 2013; Repp, 2005; Koo and Li, 2016). By grounding dexterity assessment in the structured temporal scaffold of music, RK provides a measurement tool that is both scientifically rigorous (comparable to laboratory SMS paradigms) and ecologically valid (engaging participants in a culturally meaningful activity).

Neuroscience Evidence Base, Clinical Promise, and the Measurement Gap

The use of music and rhythm in neurorehabilitation is supported by a substantial and growing body of clinical evidence demonstrating that rhythm- and music-based interventions can drive measurable improvements in motor function across diverse neurological populations, including stroke, Parkinson's disease, traumatic brain injury, and cerebral palsy (Moumdjian et al., 2017; Miendlarzewska and Trost, 2014; Braun Janzen et al., 2021). These interventions encompass a range of modalities, including Rhythmic Auditory Stimulation (RAS), Music-Supported Therapy (MST), and various forms of musical training and performance practice, each with distinct protocols, targeted outcomes, and underlying mechanisms.

Rhythmic Auditory Stimulation (RAS) for gait rehabilitation is among the most robustly evidence-based applications of rhythm in neurorehabilitation. Meta-analyses and systematic reviews have consistently demonstrated that stroke survivors and individuals with Parkinson's disease who practice walking with rhythmic auditory cues (metronome beats or rhythmic music matched to their step cadence) exhibit significant improvements in gait speed, stride length, cadence, temporal regularity, and gait symmetry compared to control groups receiving conventional therapy without auditory cues (Moumdjian et al., 2017; Thaut et al., 1999). Effect sizes are typically moderate to large (Cohen's d ranging from 0.5 to 1.0), and benefits have been shown to generalize beyond the training context and persist for weeks after cessation of cueing (Moumdjian et al., 2017). The mechanisms underlying RAS efficacy are thought to involve: (1) *entrainment* of spinal central pattern generators and cortical motor rhythms by external auditory rhythms, providing a stable temporal template that reduces variability and supports coordination (Thaut et al., 1999; Patel, 2008); (2) *scaffolding of impaired internal timing* processes by offloading temporal control onto reliable external cues, compensating for degraded basal ganglia or cerebellar timing circuits (Patel, 2008); (3) *increased saliency and predictability* of movement timing cues, which enhance attention and anticipatory motor planning (Thaut et al., 1999); and (4) *motivational and affective engagement*, which sustain practice dose and positive affect during training (Miendlarzewska and Trost, 2014; Altenmüller and Schlaug, 2015).

Extension of RAS principles to the **upper limb** has shown preliminary promise but remains less extensively studied and characterized by more heterogeneous outcomes. Several small-scale studies and pilot randomized controlled trials have demonstrated that rhythmic auditory cueing can improve reaching kinematics (movement

smoothness, timing consistency, trajectory accuracy), bimanual coordination, and upper-limb function scores in stroke survivors (Moumdjian et al., 2017; Braun Janzen et al., 2021). However, effect sizes are more variable, and it remains unclear which upper-limb tasks, cueing parameters (tempo, rhythmic complexity, modality), and patient characteristics optimize outcomes. A critical limitation is that most upper-limb RAS studies have assessed outcomes using clinical scales or gross kinematic measures (e.g., trajectory error, movement time) rather than the fine-grained timing metrics (millisecond-level asynchrony, jitter, phase correction) that are routinely quantified in SMS research (Moumdjian et al., 2017). Consequently, it is difficult to determine whether observed improvements reflect genuine enhancement of timing precision or compensatory strategies (e.g., slowing down, simplifying movements) that improve outcome metrics without addressing underlying control deficits.

Music-Supported Therapy (MST) and **musical training interventions** represent a broader class of rehabilitation approaches in which patients engage in structured musical activities—such as piano playing, percussion, electronic drum pads, or singing—as a vehicle for motor training. The rationale for MST extends beyond rhythmic entrainment to encompass multiple interrelated mechanisms (Altenmüller and Schlaug, 2015; Miendlarzewska and Trost, 2014; Patel, 2008). First, musical activities demand fine-grained motor control: piano performance, for example, requires precise timing of key presses, graded force modulation to produce dynamics, and independent finger control to articulate melodies and harmonies. These demands map directly onto the core dimensions of dexterity (timing, force scaling, individuation) that are impaired after stroke, making musical training an ecologically valid and functionally relevant practice activity. Second, music is *intrinsically motivating and emotionally engaging*, activating dopaminergic reward pathways (ventral striatum, nucleus accumbens) and limbic regions (amygdala, hippocampus) that support sustained attention, positive affect, and consolidation of motor learning (Miendlarzewska and Trost, 2014; Altenmüller and Schlaug, 2015). Neuroimaging studies have shown that listening to music, anticipating musical events, and performing music all elicit reward-related neural responses comparable to those evoked by primary rewards (food, sex) or monetary incentives, and individual differences in reward sensitivity to music predict adherence to musical training programs (Miendlarzewska and Trost, 2014). Third, musical tasks provide *immediate, intuitive auditory feedback* about movement quality: the acoustic result of a keystroke directly reflects the timing, force, and accuracy of the motor command, creating a closed sensorimotor loop that supports error detection and correction without requiring explicit instruction or external reinforcement (Altenmüller and Schlaug, 2015; Patel, 2008).

Several pilot studies and early-phase randomized controlled trials have investigated MST for upper-limb rehabilitation after stroke. For example, studies have shown that stroke survivors who practice simple piano melodies with or without augmented visual feedback (projection of target notes onto keys) exhibit improvements in finger tapping speed, Nine-Hole Peg Test scores, and Fugl-Meyer upper-limb subscale scores compared to control groups receiving conventional occupational therapy or no additional intervention (Altenmüller and Schlaug, 2015; Miendlarzewska and Trost, 2014). Effect sizes range from small to moderate (d = 0.3 to 0.7), and some studies report gains in patient-reported motivation, quality of life, and engagement with therapy (Miendlarzewska and Trost, 2014). Neurophysiological studies using EEG, transcranial magnetic stimulation (TMS), and fMRI have documented that musical training after stroke is associated with increased corticomotor excitability in the affected hemisphere, enhanced functional connectivity between motor and auditory cortices, and shifts in cortical activation patterns toward more

typical, lateralized motor representations (Altenmüller and Schlaug, 2015; Amengual et al., 2013). These neural changes suggest that musical training may promote adaptive neuroplasticity in motor networks, potentially supporting functional recovery beyond mere practice effects.

However, despite these encouraging findings, **the evidence base for music-based rehabilitation remains limited by several critical gaps**. First, sample sizes in most studies are small (often $N < 30$ per group), limiting statistical power, generalizability, and the ability to identify moderators of treatment response (e.g., lesion characteristics, impairment severity, musical background). Second, protocols are heterogeneous, varying in training dose (sessions per week, duration per session, total weeks), task content (piano, percussion, singing, rhythmic cueing), feedback modalities (auditory alone, auditory-visual, haptic), and outcome measures, making it difficult to synthesize evidence across studies or identify optimal training parameters. Third, and most critically for the present thesis, **music is typically used for its motivational or affective benefits rather than as a quantitative, precision-based testbed for measuring and targeting specific control dimensions**. Outcome measures in music-based intervention studies are usually limited to clinical scales (ordinal scores on Fugl-Meyer, ARAT, Box and Block), gross performance metrics (number of notes played, time to complete a melody, overall accuracy), or self-reported motivation and satisfaction (Altenmüller and Schlaug, 2015; Miendlarzewska and Trost, 2014). These measures capture global improvements or motivational engagement but do not quantify the fine-grained timing precision (millisecond-level asynchrony, jitter, phase correction), force steadiness (coefficient of variation, smoothness), or individuation (co-contraction, motor overflow) that are the hallmarks of skilled dexterity and that likely underlie functional gains. Without high-resolution, continuous measures of these control dimensions, it is impossible to determine whether musical training improves precision in specific domains, how those improvements relate to neuromuscular reorganization, or whether gains generalize to non-musical functional tasks.

This **measurement gap** is not merely a matter of experimental convenience or rigor; it has direct implications for clinical translation and mechanistic understanding. If music-based interventions are to transition from pilot studies to large-scale, definitive clinical trials, they require validated, reliable, sensitive outcome measures that can detect meaningful change, track dose-response relationships, and serve as surrogate markers of functional recovery. Moreover, if the goal is to understand *why* musical training works—what specific neuromuscular, cognitive, or motivational mechanisms drive observed benefits—then it is essential to measure the control processes (timing, force, individuation) directly rather than inferring them from distal outcome measures. Without such measures, the field risks remaining in a state of empirical trial-and-error, testing interventions without a clear mechanistic rationale or the ability to optimize protocols systematically.

The **Rhythm Karaoke (RK) project** addresses this measurement gap by developing and psychometrically validating a music-based assessment protocol that quantifies timing precision at millisecond resolution using the Dextrain Finger Force Manipulandum. RK operationalizes music not as a generic motivational tool but as a *quantitative experimental paradigm* in which structured temporal demands, systematic feedback manipulations, and high-fidelity force-time recording converge to reveal the specific dimensions of sensorimotor control that are impaired after stroke and that may be targeted by rehabilitation interventions. By doing so, RK fills the measurement gap, providing a tool that bridges the ecological validity and motivational appeal of music with the precision and psychometric rigor of laboratory

SMS research. This positions music-based assessment as a complement to clinical scales and a potential surrogate outcome for trials of music-based or timing-focused interventions.

Beyond the motivational and timing-scaffolding mechanisms discussed above, music also offers a unique opportunity to investigate the role of *reward, semantic content, and emotional engagement* in motor learning and adherence. Recent neuroscience research has demonstrated that the brain's reward system responds not only to the achievement of goals (extrinsic rewards) but also to the intrinsic pleasure and aesthetic satisfaction derived from music (Miendlarzewska and Trost, 2014). Functional MRI studies show that listening to pleasurable music activates the ventral striatum, a core component of the dopaminergic reward circuit, and that the degree of activation predicts both subjective reports of pleasure and physiological responses (skin conductance, heart rate) (Miendlarzewska and Trost, 2014). Individual differences in musical reward sensitivity (musicality, hedonic capacity) predict engagement with musical tasks, practice persistence, and ultimately motor learning outcomes in training studies (Miendlarzewska and Trost, 2014). This suggests that for a subset of individuals, music-based training may leverage intrinsic reward processes to sustain high-dose, high-quality practice in ways that gamification or extrinsic incentives cannot replicate.

Furthermore, preliminary evidence from the **Rhythm Karaoke project** suggests that the *semantic and melodic content* of auditory stimuli modulates both timing performance and subjective engagement. In pilot data, participants performing rhythmic tapping synchronized to speech-like stimuli generated by the Voks vocal synthesizer (Locqueville et al., 2020) exhibited different timing profiles (asynchrony, variability) and reported different levels of engagement compared to tapping to abstract rhythmic tones or metronome beats. This finding aligns with theoretical predictions from the OPERA hypothesis (Patel, 2008), which posits that auditory-motor coupling is enhanced when auditory stimuli are meaningful, emotionally salient, and processed through multimodal cortical networks (involving language, semantic, and affective regions in addition to motor cortex). If semantic-melodic content indeed modulates timing precision and engagement, then the design of music-based interventions should not treat auditory cues as interchangeable metronomes but should carefully consider stimulus content, cultural relevance, and individual preferences. The RK project is uniquely positioned to investigate these questions by systematically varying stimulus content (speech-like Voks syllables, melodic fragments, rhythmic tones, metronome) and measuring both objective timing metrics and subjective engagement, thereby informing evidence-based design principles for rehabilitation interfaces.

In summary, the neuroscience evidence base for rhythm- and music-based interventions is substantial and growing, demonstrating that musical activities can drive motor improvements, engage reward and timing networks, and support neuroplastic reorganization. However, a critical gap remains: music is rarely deployed as a *quantitative precision testbed* with the measurement resolution, experimental control, and psychometric validation necessary to serve as a reliable outcome measure or mechanistic probe. The RK project fills this gap by positioning music as both an engaging activity and a rigorous experimental paradigm, thereby advancing the field from demonstration of feasibility and preliminary efficacy toward mechanistic understanding and precision-based intervention design.

Auditory Feedback as an Experimental and Clinical Lever: Probing Internal Control and Informing Design

The third and final pillar of the scientific rationale for music as a precision testbed concerns the role of *auditory feedback* as an experimental lever for investigating internal control processes and as a design element for rehabilitation interfaces. In everyday motor behavior, sensory feedback—visual, proprioceptive, tactile, and auditory—provides critical information about the state of the body and the consequences of movements, enabling online error correction, adaptive adjustments to changing task demands, and the refinement of internal models that support anticipatory control (Wong et al., 2019; Patel, 2008). Auditory feedback, in particular, occupies a unique niche in motor control: while vision and proprioception provide direct information about limb position and contact forces, audition provides temporal information about the timing and sequencing of events, which is especially relevant for rhythmic, sequential, or speech-related motor tasks (Patel, 2008; Zatorre, Chen, and Penhune, 2007).

Musical performance is perhaps the paradigmatic domain in which auditory feedback plays a central, even indispensable role. Pianists, for example, rely on the sound of each keystroke to confirm that the intended note was produced, to assess whether the timing and dynamics were appropriate, and to adjust subsequent movements accordingly (Patel, 2008). This dependence on auditory feedback has been exploited in motor control research through paradigms involving **auditory feedback manipulations**—systematically altering, delaying, or removing the auditory consequences of actions to probe the sensorimotor control architecture and its reliance on different sources of information (Repp and Su, 2013).

One well-studied manipulation is **Delayed Auditory Feedback (DAF)**, in which the sound produced by a movement is played back to the performer with a temporal delay (typically 50-300 milliseconds) (Repp and Su, 2013; Hardy et al., 2018). DAF has been used extensively in speech research, where it is known to disrupt fluency, induce stuttering-like dysfluencies, slow speech rate, and increase error rates even in neurologically intact speakers (Hardy et al., 2018). The disruptive effects of DAF are thought to arise from a conflict between the expected timing of auditory reafference (based on the efference copy of the motor command and an internal forward model of the vocal tract) and the actual timing of the delayed auditory signal, leading to error signals that trigger corrective adjustments that are themselves inappropriate and thus further degrade performance (Hardy et al., 2018; Patel, 2008). Interestingly, the magnitude of DAF disruption varies across individuals and clinical populations: individuals with cerebellar lesions, for example, show reduced sensitivity to DAF, suggesting that the cerebellum plays a critical role in computing and responding to sensory prediction errors (Hardy et al., 2018). In clinical contexts, DAF sensitivity has been investigated as a potential diagnostic or therapeutic probe for speech and motor disorders, including primary progressive aphasias and Parkinson's disease (Hardy et al., 2018).

In the context of **music-based rhythmic synchronization**, DAF provides a powerful tool for investigating the relative contributions of feedforward (predictive, internal-model-based) and feedback (reactive, error-correction-based) control to timing precision (Repp and Su, 2013). If auditory feedback is delayed, participants must rely more heavily on proprioceptive and tactile cues or on internal timekeeping to maintain synchronization, and the degradation in performance (increased asynchrony, increased variability, reduced error correction) reveals the extent to which auditory reafference was supporting timing control in the unperturbed condition. Early SMS

research demonstrated that moderate DAF (on the order of 100-150 ms) substantially increases asynchrony variability and disrupts phase correction, while very short delays (<50 ms) or very long delays (>250 ms) have less impact, consistent with the hypothesis that auditory feedback is most critical for online error correction at timescales where sensory delays match the natural dynamics of corrective motor adjustments (Repp and Su, 2013).

The **Rhythm Karaoke (RK) project** systematically exploits auditory feedback manipulations—including synchronous feedback, delayed feedback, altered accent patterns, and removal of feedback—to probe the sensorimotor control strategies employed by stroke survivors and healthy controls during rhythmic finger tapping. By comparing timing metrics (asynchrony, variability, lag-1 autocorrelation) across feedback conditions, RK can quantify the extent to which individuals rely on auditory reafference vs. internal timing, identify deficits in adaptive error correction that may be masked in normal feedback conditions, and reveal individual differences in control strategies that may predict responsiveness to different types of training or feedback interventions. For example, if a stroke survivor exhibits disproportionately large increases in timing variability when auditory feedback is delayed or removed, this suggests that their internal timekeeping or proprioceptive-based control is compromised, and that rehabilitation interventions should emphasize strengthening these internal processes (e.g., through practicing without auditory cues or with only sparse, intermittent feedback) rather than relying solely on external auditory scaffolding (Repp and Su, 2013).

Beyond DAF, another important dimension of auditory feedback manipulation concerns the **temporal microstructure and accent patterns** of rhythmic stimuli. In music and speech, not all temporal events are perceptually equivalent: certain events are perceived as "accented" (more salient, structurally important) due to differences in intensity, pitch, duration, or timbral properties, and these accents shape the listener's interpretation of metrical structure and influence the timing of synchronized movements (Palmer and Krumhansl, 1990; Repp and Su, 2013). For instance, in a simple 4/4 meter, the first beat of each measure (the downbeat) is typically perceived as stronger than the other beats, and listeners and performers naturally align their movements more precisely to downbeats than to weak beats (Palmer and Krumhansl, 1990). This phenomenon, known as the **accent-induced bias** or **hierarchical timing effect**, demonstrates that perceptual interpretation of rhythmic structure influences motor timing, and that the brain does not simply respond to the raw temporal pattern of auditory events but constructs a higher-level representation of meter and grouping that guides motor planning and execution (Palmer and Krumhansl, 1990; Patel, 2008).

A related concept with direct implications for music-based rehabilitation design is the **perceptual center (P-center)** of a syllable or musical note (Marcus, 1981). The P-center is the moment within the acoustic signal that is perceived as the "beat" or "onset" of the sound—the temporal anchor point to which listeners and performers synchronize their movements. Critically, the P-center does not necessarily coincide with the physical acoustic onset of the sound (the moment when sound energy first exceeds baseline) but is typically located slightly later, at or near the moment of maximum perceptual salience, which depends on the sound's amplitude envelope, spectral properties, and phonetic structure (Marcus, 1981). For speech syllables, the P-center is typically located 20-50 milliseconds after the acoustic onset, closer to the vowel nucleus than to the consonantal onset (Marcus, 1981). If a rehabilitation interface or training protocol presents auditory cues to guide finger movements, and if those cues are speech-like or have complex amplitude envelopes, then failing to

account for the P-center offset can introduce systematic timing biases: users may synchronize their movements to the perceptual beat (P-center) rather than to the physical onset, leading to apparent "errors" that are in fact perceptually correct but physically misaligned (Marcus, 1981).

The **Rhythm Karaoke project**, in collaboration with researchers developing the **Voks vocal synthesizer** (Locqueville et al., 2020), has specifically addressed this issue by using Voks-generated speech-like syllables as rhythmic cues and by applying a temporal offset correction (approximately 30 milliseconds) to align the target beat with the estimated P-center rather than the acoustic onset. Pilot data from the RK project suggest that this P-center correction improves synchronization accuracy (reduces mean asynchrony) and subjective ease of synchronization, and that participants are more engaged and report that the task "feels more natural" when cues are speech-like (Voks syllables with semantic content) compared to abstract tones or clicks (Locqueville et al., 2020). These findings have direct design implications for rehabilitation interfaces: if the goal is to support precise timing and sustained engagement, then auditory cues should be designed with attention to perceptual salience, P-center alignment, semantic-melodic content, and cultural relevance, rather than treating auditory feedback as a simple acoustic "beep" that cues movement onset.

Furthermore, the interplay between auditory feedback content and motor control extends to **multimodal integration** and the construction of internal models. According to predictive coding and internal model frameworks (Patel, 2008; Wong et al., 2019), the brain continuously generates predictions about the sensory consequences of actions (auditory, visual, proprioceptive) based on efference copies of motor commands and learned forward models of the body and environment. When actual sensory feedback matches predictions, the system runs smoothly with minimal conscious attention; when mismatches occur (prediction errors), these errors are propagated through the network to update internal models and adjust subsequent motor commands (Wong et al., 2019). Auditory feedback plays a critical role in this predictive loop for musical and speech tasks: the brain predicts the sound that will result from a finger press or vocal articulation, compares this prediction to the actual auditory input, and uses the error signal to refine timing, force, or articulatory parameters on the next trial (Patel, 2008; Hardy et al., 2018).

Stroke can disrupt this predictive sensorimotor loop at multiple levels: lesions affecting sensory cortices impair the processing of afferent signals; lesions affecting parietal or cerebellar regions impair the construction or updating of forward models; and lesions affecting motor cortex or basal ganglia impair the use of prediction errors to adjust motor commands (Wong et al., 2019; Cramer et al., 2011). By systematically manipulating auditory feedback (introducing delays, removing feedback, altering content), the RK project can probe which components of the predictive loop are preserved or compromised in individual stroke survivors, thereby informing the design of personalized rehabilitation interventions that compensate for specific deficits (e.g., providing more salient, redundant feedback for individuals with sensory processing impairments) or that target specific processes for retraining (e.g., practicing with degraded or absent feedback to strengthen internal models for individuals whose feedback-dependent control is excessive or maladaptive) (Repp and Su, 2013; Wong et al., 2019).

Finally, the investigation of auditory feedback as an experimental and clinical lever extends beyond timing to encompass **force modulation and haptic-auditory integration**. In musical contexts, force (keystroke velocity in piano, bow pressure in strings) directly determines the loudness and timbre of the sound produced, creating a tight coupling between kinetic output and auditory feedback (Patel, 2008).

This coupling can be exploited in rehabilitation by mapping force to salient auditory dimensions (pitch, volume, timbre) to provide intuitive, continuous feedback about force control quality. Preliminary work in the RK and ReTouche projects has explored such mappings, for example by having participants modulate keystroke force to control musical dynamics or by providing auditory "error tones" when force exceeds or falls below target ranges. Participants report that auditory force feedback is intuitive and supports attention to force steadiness and scaling in ways that visual displays alone do not, likely because auditory feedback is processed more rapidly and requires less overt attentional resources than vision (Patel, 2008). Moreover, auditory feedback can be spatially non-localized (heard through headphones or speakers without requiring gaze direction), allowing users to maintain visual attention on their hand or on task-relevant visual cues while still receiving rich sensory information about performance quality.

The **HD-EMG + AI project**, which decodes continuous finger forces from muscle activity, can be integrated with RK and ReTouche to create *EMG-driven auditory biofeedback systems* in which decoded forces (rather than measured forces from the Dextrain) control musical output. An exploratory application of particular interest is the **EMG-to-Disklavier interface**, in which decoded muscle activations drive a robotic piano (Yamaha Disklavier) to produce musical notes in real time. This creates a closed sensorimotor loop in which users "hear" their muscle activity as music, providing immediate, interpretable feedback about the timing, smoothness, and independence of muscle recruitment without requiring conscious interpretation of raw EMG traces (which are opaque even to expert clinicians) (Miendlarzewska and Trost, 2014; Altenmüller and Schlaug, 2015). Such interfaces leverage the motivational and aesthetic appeal of music while providing precise, moment-to-moment information about neuromuscular control quality, positioning them as tools for both motor learning and engagement.

In summary, auditory feedback serves multiple roles in music-based rehabilitation: it is an *experimental probe* for investigating sensorimotor control architecture and predictive coding processes; it is a *design element* that can be systematically manipulated (timing, content, salience, mapping) to optimize support for timing, force, and error correction; and it is an *intrinsically motivating channel* through which performance quality is rendered audible and aesthetically engaging. The RK, HD-EMG + AI, and ReTouche projects collectively exploit these affordances, positioning auditory feedback not as a peripheral or incidental feature but as a central, theoretically grounded, and experimentally manipulable component of the rehabilitation paradigm.

Synthesis and Transition: Music as the Ideal Convergence Point for the Tripartite Research Program

The preceding discussion has established that music provides a unique and scientifically justified context for investigating fine manual dexterity and designing precision-based rehabilitation interventions. To synthesize: (1) music imposes *structured temporal demands* with hierarchical rhythmic organization that afford controlled experimental manipulation, engage distributed sensorimotor networks, and demand millisecond-level timing precision measurable with objective, continuous metrics; (2) the *neuroscience evidence base* demonstrates that musical activities engage auditory-motor coupling, support neuroplastic reorganization, and activate reward circuits, yet a critical measurement gap remains because music is rarely deployed as a quantitative precision testbed with validated, high-resolution outcome measures; and (3)

auditory feedback serves as a powerful experimental lever for probing internal control processes (through manipulations such as DAF, accent alteration, and P-center adjustment) and as a design element that enhances error awareness, supports force modulation, and sustains engagement through aesthetic and semantic salience.

These three pillars converge to position music as the ideal testbed for the **tripartite research architecture** of this thesis:

- **For Measurement (RK):** Music provides the structured temporal scaffold necessary for high-resolution quantification of timing precision, the ecological validity needed for clinical relevance, and the experimental control needed for psychometric validation and sensitivity to feedback manipulations. The RK project operationalizes music not as a vague motivational adjunct but as a rigorous SMS paradigm integrated with the Dextrain Manipulandum, yielding reliable, millisecond-resolution timing metrics that **resolve Gap 1** (lack of validated precision measures).

- **For Mechanism (HD-EMG + AI):** Music provides a functional, goal-directed context in which timing, force, and individuation are all task-relevant simultaneously, allowing HD-EMG recordings and AI-driven force decoding to be linked to behaviorally meaningful performance metrics. By recording 224-channel HD-EMG during RK tasks and decoding continuous finger forces in real time, the HD-EMG + AI project reveals the neuromuscular substrates (co-contraction, synergies, neural drive dynamics) underlying timing precision and provides mechanistic explanations for behavioral deficits, thereby **addressing Gap 2** (lack of mechanistic understanding).

- **For Engagement (ReTouche):** Music provides the intrinsic motivation, aesthetic satisfaction, and cultural meaningfulness that sustain long-term practice, while also affording multimodal feedback (auditory, visual, haptic) that can be designed to support attention, error awareness, and strategic learning. The ReTouche project investigates how interface and feedback design shape the lived experience of musical learning, generating evidence-based design principles that ensure RK and HD-EMG + AI tools are not only scientifically rigorous but also engaging, usable, and effective in real-world practice, thereby **ensuring sustained engagement** and **defining design principles to resolve Gap 3** (lack of engagement-centered design).

In sum, music is not merely a convenient or appealing choice for this research program; it is the *scientifically optimal convergence point* where measurement precision, mechanistic depth, and engagement sustainability meet. The structured temporal demands of music enable quantitative rigor; the neuroscience evidence base and reward mechanisms support motivation and plasticity; and the auditory feedback affordances provide both experimental control and design flexibility. By positioning music as a precision testbed rather than a motivational add-on, this thesis advances rehabilitation science toward a future in which fine dexterity is measured at millisecond and sub-Newton resolution, understood at the neuromuscular level through AI-decoded EMG, and trained through engaging, user-centered interfaces that respect the lived experience and agency of individuals working to rebuild skilled hand function after stroke.

The next section will present the three projects—Rhythm Karaoke (RK), HD-EMG + AI, and ReTouche—in detail, showing how each operationalizes the rationale developed here and how together they constitute a coherent, integrated research

program addressing the three fundamental gaps identified in the state-of-the-art review.

1.3 Our Approach: Three Complementary Lenses

The three fundamental gaps identified in the preceding review—**[Gap 1]** lack of reliable, high-resolution measurement tools for timing precision at millisecond level; **[Gap 2]** lack of mechanistic understanding linking behavioral deficits to neuromuscular processes in real-time; and **[Gap 3]** lack of engagement-centered design principles grounded in lived experience—constitute the **central research questions** that structure this thesis. These gaps are not merely limitations of prior work; they represent unresolved scientific challenges that demand novel methodological approaches and integrated solutions. The tripartite architecture of this thesis directly addresses these research questions through three complementary projects, each operationalizing a distinct lens (Measurement, Mechanism, Engagement) while maintaining music as a unifying, ecologically valid, and *quantitative assessment framework*. Table 1.3 below synthesizes the core objectives, methods, quantitative capabilities, gap-filling contributions, and status of each project, clarifying how they collectively form a coherent research program designed to answer these fundamental questions.

Table 1.3: Thesis Architecture — The Measurement-Mechanism-Engagement Triad

TABLE 1.3: The three projects of this thesis, their objectives, methods with quantitative specifications, gap-filling contributions, and status.

Project	Principal Objective	Methods / Key Capabilities	Contribution to Addressing Gaps	Status & ences
1. Rhythm Karaoke (RK) *Measurement Lens*	**Measure:** Establish a reliable, sensitive measure of timing precision for fine finger control.	Rhythmic tapping on Dextrain Manipulandum (**0.01 N** resolution) synchronized to music/speech stimuli. Psychometric validation: test-retest reliability (ICC) (Koo and Li, 2016), SEM, MDC. Systematic auditory feedback manipulations (synchronous, delayed, altered accents).	**Resolves Gap 1 [RK]:** Provides high-resolution millisecond-level timing metrics (asynchrony, variability, error correction, phase drift) unavailable in MST or standard clinical assessments. Enables quantitative precision assessment for research and clinical trials.	Manuscri (2025, in (Arslan, berg, 202 al., 2015)
2. HD-EMG + AI *Mechanism Lens*	**Model:** Continuous proportional decoding of finger forces (multi-DOF) from **224-channel** HD-EMG to reveal neuromuscular mechanisms.	AI regression (CNN-LSTM) on **224 electrodes** (including intrinsic hand muscles: thenar, hypothenar, interossei, lumbricals). Target: $R^2 > 0.75$; Latency target **<50 ms**. Cross-user generalization without fine-tuning. Synchronized with Dextrain for ground-truth force.	**Models Gap 2 [HD-EMG]:** Provides mechanistic account linking behavioral performance (RK) to underlying neuromuscular substrates (co-contraction, neural drive, synergies). Real-time interpretable biofeedback. Includes intrinsic hand muscles (rarely recorded).	Manuscri (2024, in (confirme et al., 20 Farina, 2

Project	Principal Objective	Methods / Key Capabilities	Contribution to Addressing Gaps	Sta enc
3. ReTouche *Engagement Lens*	**Engagement:** Define HCI design principles supporting attention, agency, and sustained practice of fine motor control.	Qualitative studies: Autoethnography ($N = 3$, 2-week piano learning journal), CSO ($N = 18$), Focus Groups ($N = 4$). Embodied interaction framework: projection mapping (timing/force visualization), performance replay (Kirsh, 2013).	**Ensures sustained engagement addressing Gap 3 [ReTouche]:** Provides design principles ensuring long-term adherence and experience quality, crucial for chronic rehabilitation. Renders RK/HD-EMG tools usable at home. Reveals how feedback externalizes errors, reorients attention from outcomes to processes.	Ma mo rev (Au mer Tak

This tripartite architecture is conceptually coherent: **measurement (RK) provides the reliable behavioral metrics** that quantify precision and enable rigorous evaluation of interventions; **mechanism (HD-EMG + AI) provides the explanatory power** that links observable behavior to underlying neuromuscular processes, informing the design of targeted, feedback-driven interventions; and **engagement (ReTouche) provides the human-centered perspective** that ensures tools and protocols are not only effective in controlled settings but also usable, motivating, and sustainable in real-world practice. Music threads through all three lenses as a precision-demanding, ecologically valid, and intrinsically rewarding activity, bridging the gap between laboratory psychophysics and clinically relevant rehabilitation.

The following subsections present each project in greater detail, outlining the specific scientific rationale, methodological approaches with quantitative specifications, anticipated contributions to filling the identified gaps, and integration points that demonstrate how the three projects constitute a coherent, mutually reinforcing research program. Each subsection is structured to progress from conceptual motivation through technical implementation to expected impact, with particular attention to the quantitative rigor (measurement precision, decoding performance targets, sample sizes) that distinguishes this approach from previous work and ensures clinical translatability.

1.3.1 Rhythm Karaoke (RK): The Measurement Lens—Validated Timing Precision Assessment for Fine Manual Dexterity

Gap Addressed: Conventional clinical assessments (Box & Block Test, Nine-Hole Peg Test, Fugl-Meyer) lack the millisecond-level temporal resolution needed to quantify fine timing precision during functional manual tasks. Music-based interventions (RAS, MST) have shown promise, yet outcome measures rarely capture the temporal dynamics of auditory-motor synchronization central to proposed mechanisms (Térémetz et al., 2015; Moumdjian et al., 2017).

Instrumentation & Task Design: The Rhythm Karaoke (RK) project integrates the **Dextrain Finger Force Manipulandum** (0.01 N force resolution, 1 kHz sampling) with music-based sensorimotor synchronization (SMS) paradigms (Térémetz et al., 2015). Participants synchronize rhythmic finger presses to auditory stimuli ranging from simple isochronous metronomes to musical sequences with hierarchical meter and speech-like syllable sequences generated using the **Voks vocal synthesizer** with controlled P-center alignment (Locqueville et al., 2020; Marcus, 1981). This diversity allows assessment of basic synchronization capacity and the ability to exploit hierarchical temporal structure and perceptually salient events (accents, P-centers) (Palmer and Krumhansl, 1990; Patel, 2008).

Behavioral Metrics: The high-resolution timing data afford computation of validated SMS metrics (Repp, 2005; Repp and Su, 2013): (1) **Mean Signed Asynchrony (MSA)**, the average temporal offset indicating anticipatory (-20 to -50 ms, feedforward control) vs. reactive (positive, feedback-driven) timing; (2) **Asynchrony Variability (SD)**, reflecting consistency and stability of synchronization; (3) **Lag-1 Autocorrelation**, quantifying error correction strength (typical values -0.3 to -0.5 indicate beat-by-beat adjustment); and (4) **Phase Drift**, the systematic shift in mean asynchrony over time, indicating tempo representation stability.

Auditory Feedback Manipulations as Experimental Probes: RK incorporates systematic auditory feedback manipulations serving as experimental probes of sensorimoor control mechanisms: (1) **Delayed Auditory Feedback (DAF)** at 50-300 ms delays to dissociate feedforward and feedback contributions; (2) **Accent Structure Manipulation** (removing or altering amplitude/pitch accents on downbeats) to assess reliance on hierarchical metrical cues vs. isochronous timing; (3) **P-center Alignment** using Voks-synthesized speech-like syllables with $\sim$30 ms offset aligning acoustic onset with perceptual center, predicted to reduce asynchrony variability by 15-25% (Locqueville et al., 2020; Marcus, 1981); and (4) **Semantic and Emotional Content Modulation** (familiar/unfamiliar, positive/neutral

valence), predicted to modulate timing precision and error correction via attention and motivation (Miendlarzewska and Trost, 2014).

Psychometric Validation: RK validation follows rigorous best-practice guidelines (Koo and Li, 2016): **Test-retest reliability** assessed via Intraclass Correlation Coefficient (ICC > 0.75 target for asynchrony variability, ICC > 0.70 for secondary metrics), Standard Error of Measurement (SEM), and Minimal Detectable Change (MDC) computed across two sessions separated by 7-14 days in healthy participants ($N \geq 30$) and chronic stroke survivors ($N \geq 30$). **Construct validity** evaluated through: (a) *convergent validity*—moderate correlations (Pearson $r = 0.40$-0.70) with established clinical measures (Fugl-Meyer Upper Extremity, Box and Block Test) and Dextrain static force metrics; (b) *discriminant validity*—significant differences between stroke survivors and age-matched controls (Cohen's $d > 0.80$ for asynchrony variability); and (c) *sensitivity to experimental manipulations*—predicted effects of DAF (increased variability, reduced error correction), P-center alignment (15-25% variability reduction), and valence modulation (familiar/positive stimuli improve precision).

Expected Contribution: RK provides (1) the first **validated, millisecond-resolution timing precision assessment** for stroke rehabilitation grounded in SMS theory and meeting psychometric standards (ICC, SEM, MDC); (2) **ecological validity** via musical/speech stimuli that engage attention and motivation while affording experimental control; (3) **mechanistic profiling** distinguishing feedforward vs. feedback-driven control strategies through systematic feedback manipulations; and (4) a rigorous **outcome measure for clinical trials** evaluating rhythm-based interventions. By resolving Gap 1, RK establishes the quantitative measurement foundation upon which mechanistic understanding (HD-EMG + AI) and engagement-centered design (ReTouche) are built. **Detailed methods, validation results, and clinical applications are presented in Chapter 3.**

1.3.2 HD-EMG + AI: The Mechanism Lens—Decoding Continuous Finger Forces from High-Density Neuromuscular Signals to Reveal Control Substrates

Gap Addressed: While RK quantifies behavioral precision, it does not reveal the neuromuscular substrates underlying timing and force control. High-density electromyography (HD-EMG) offers rich spatiotemporal patterns of muscle activation, but the sheer dimensionality (224 channels) and complexity of raw signals create an **interpretability barrier**: clinicians cannot extract actionable insights, patients cannot use raw EMG for real-time biofeedback, and researchers lack tools to mechanistically link behavioral metrics (RK asynchrony, variability) to underlying neuromuscular processes (co-contraction, individuation, motor unit recruitment) (Farina et al., 2010; Muceli and Farina, 2012).

Instrumentation & AI Decoding Pipeline: The HD-EMG + AI project addresses this gap by integrating a **224-channel HD-EMG system** covering forearm extrinsic muscles (flexor digit or profundus, extensor digitorum) and, critically, intrinsic hand muscles (thenar, hypothenar, interossei, lumbricals)—regions rarely recorded due to placement challenges but essential for finger individuation (Muceli and Farina, 2012). The system synchronizes with the Dextrain Manipulandum (1 kHz force, 2048 Hz EMG) to provide ground-truth force labels for supervised learning. **AI regression models** (Convolutional Neural Networks [CNNs], Long Short-Term Memory networks [LSTMs], CNN-LSTM hybrids, and exploratory transformer architectures) are trained to decode continuous, multi-degree-of-freedom (multi-DOF) finger forces from high-dimensional EMG input (Montazerin et al., 2023; Farina et al., 2010). Performance targets are stringent: $R^2 > 0.75$ (Pearson $r > 0.87$, threshold for clinical interpretability), **latency <50 ms** (preserving sense of agency for real-time biofeedback), and **cross-user generalization** (models trained on pooled data achieve $R^2 \geq 0.70$ on held-out participants, reducing calibration burden) (Farina et al., 2010).

Mechanistic Insight & Clinical Value: Decoded finger forces provide three complementary values: (1) **Interpretable clinical readouts**—clinicians view predicted force trajectories, individuation indices, and co-contraction metrics rather than 224 inscrutable waveforms; (2) **Real-time biofeedback**—decoded forces drive visual displays (force curves, virtual hand animations), auditory mappings (force controlling pitch/timbre), or haptic cues, guiding patients toward efficient activation without requiring EMG expertise; (3) **Behavior-mechanism linkage**—by recording HD-EMG during RK tasks, the project links millisecond-level behavioral precision (asynchrony variability, error correction) to neuromuscular patterns (spatial selectivity, temporal modulation, synergy structure), enabling mechanistic profiling: Do individuals with weak error correction exhibit delayed EMG modulation? Do high-variability performers show excessive co-contraction or reduced intrinsic muscle recruitment? Such insights inform targeted interventions (e.g., biofeedback highlighting co-contraction for individuals with excessive antagonist activation) (Farina et al., 2010; Muceli and Farina, 2012).

EMG-to-Disklavier Musical Biofeedback: An exploratory application demonstrates proof-of-concept for engaging, musically grounded biofeedback: decoded finger forces in real time drive a **Yamaha Disklavier** (robotic acoustic piano), creating a closed sensorimotor loop in which users "hear"

their neuromuscular control as musical output (pitch, loudness, timbre mapped to force amplitude, rate, individuation). This interface leverages music's motivational and aesthetic power while providing immediate, ecologically valid feedback about activation quality, positioning HD-EMG biofeedback not merely as a clinical tool but as an engaging interface supporting intrinsic motivation (Miendlarzewska and Trost, 2014).

Expected Contribution: HD-EMG + AI provides (1) state-of-the-art AI decoding meeting clinical performance thresholds ($R^2 > 0.75$, <50 ms latency, cross-user generalization); (2) interpretable neuromuscular readouts overcoming HD-EMG's opacity; (3) mechanistic links between RK behavioral metrics and underlying control processes; and (4) proof-of-concept for musically engaging biofeedback (EMG-to-Disklavier). By resolving Gap 2, HD-EMG + AI enables rehabilitation to move from descriptive assessment toward mechanistic understanding and deficit-specific, personalized biofeedback protocols. **Detailed methods, decoding architectures, validation results, and clinical applications are presented in Chapter 4.**

1.3.3 ReTouche: The Engagement Lens—Defining Human-Centered Design Principles for Sustained Practice and Agency in Fine Motor Learning

Gap Addressed: Quantitative assessment (RK) and mechanistic insight (HD-EMG + AI) are necessary but insufficient if users do not engage consistently with rehabilitation tools over the weeks and months required for motor recovery. Conventional rehabilitation research prioritizes efficacy (does the intervention work in controlled settings?) while neglecting **engagement and adherence**—the human factors that determine whether tools are adopted, used regularly, and sustained in real-world practice. This gap is particularly acute for technology-mediated, home-based rehabilitation, where attrition rates exceed 50% and lack of motivation, unclear progress feedback, and disconnection from meaningful goals are primary barriers (Kirsh, 2013). The third gap, therefore, is the lack of evidence-based design principles grounded in the lived experience of users that can inform the development of rehabilitation interfaces supporting sustained engagement, agency, and intrinsic motivation.

Qualitative Methods & Theoretical Framework: ReTouche addresses this gap through qualitative, phenomenologically oriented research investigating the lived experience of learning fine manual coordination in a musical context—an analog for rehabilitation that shares key demands (precise timing, force modulation, sustained practice, gradual skill building) while offering ecological validity and cultural meaningfulness (Kirsh, 2013; Klemmer, Hartmann, and Takayama, 2006). The study employs **mixed qualitative methods** with methodological triangulation to ensure rigor: (1) **Longitudinal Autoethnography** ($N = 3$ participants, 2-week daily piano practice journals) capturing first-person experiential depth, strategic adaptations, and motivational fluctuations; (2) **Comparative Structured Observation (CSO)** ($N = 18$ novice piano learners) with counterbalanced conditions (baseline minimal feedback vs. intermediate visual-only vs. full ReTouche multimodal) to systematically compare interface affordances and subjective experience (post-session questionnaires, semi-structured interviews); and (3) **Expert Focus Groups** ($N = 4$ occupational therapists, rehabilitation robotics researchers) for external validation and clinical translation insights. The **Embodied Interaction** theoretical framework (Kirsh, 2013; Klemmer, Hartmann, and Takayama, 2006) guides analysis: skilled behavior emerges through iterative perception-action coupling shaped by tool affordances, and interfaces should externalize internal states (errors, goals), support fluent action-perception loops, and respect users' sense of agency and meaning-making.

System Design & Interface Affordances: The ReTouche platform centers on an augmented **Yamaha Disklavier** (robotic acoustic piano) integrated with custom software providing: (1) **Projection Mapping**—spatial visual cues (colored lights, timing bars, force intensity indicators) projected directly onto piano keys and surrounding surfaces, creating embodied, spatially situated representations rather than abstract screen displays; (2) **Multimodal Feedback**—auditory (acoustic piano sound, metronome, verbal encouragement), visual (projection animations showing timing accuracy, force quality, note correctness), and haptic guidance (moving keys demonstrating rhythm and timing, fading as skill develops); (3) **Scaffolding Features**—tempo control (users slow down difficult passages), segmentation (practice focus on specific measures or hands separately), performance replay (immediate playback with annotated visualization showing which notes were early/late/on-time, force overshoots), and longitudinal tracking (session-by-session progress graphs). These affordances operationalize design hypotheses: error externalization supports strategic correction, multimodal feedback directs attention to movement quality, agency-supporting features sustain motivation, and adaptive scaffolding promotes independence (Kirsh, 2013).

Key Findings & Design Principles: Thematic analysis of autoethnography journals, CSO interviews, and focus group transcripts revealed six evidence-based design principles:

(1) **Externalize Errors Through Persistent, Inspectable Representations:** Participants consistently reported that replay and timeline visualizations transformed error detection from implicit (relying on degraded real-time auditory feedback) into explicit, reflective processes supporting strategic correction. This principle suggests rehabilitation interfaces should provide post-hoc review capabilities, not only immediate feedback (Kirsh, 2013).

(2) **Reorient Attention from Outcomes to Processes:** Baseline conditions elicited outcome-focused attention ("Did I play the right notes?"), while full ReTouche conditions shifted attention to movement quality ("How smooth/well-timed were my movements?"). Process-oriented attention supports deeper, more transferable motor learning (Kirsh, 2013).

(3) **Support Agency Through Meaningful Choices and Immediate Effects:** Sustained engagement over 2 weeks (80% adherence in autoethnography) was attributed to sense of agency: tempo control, segmentation choice, and immediate contingent feedback preserved users' perception of control and self-determination (Kirsh, 2013).

(4) **Make Progress Visible and Salient:** Replay comparisons showing session-to-session improvement, successful completion of progressively difficult segments, and subjective feelings of fluency were consistently cited as motivators, suggesting interfaces should track and visualize progress longitudinally (Kirsh, 2013).

(5) **Leverage Intrinsic Motivation Through Culturally Meaningful Activities:** Music's aesthetic satisfaction, social connection (duet-like interaction with Disklavier), and personal meaningfulness (selecting familiar melodies) sustained motivation where abstract motor tasks might not, underscoring importance of embedding rehabilitation in culturally meaningful contexts (Miendlarzewska and Trost, 2014).

(6) **Fade Scaffolding Gradually to Promote Independence:** Haptic guidance (moving keys) was valued early but became a "crutch" when prolonged, pointing to need for adaptive fading: rich scaffolding initially, progressive reduction to foster feedback-independent control (Kirsh, 2013).

Expected Contribution: ReTouche provides (1) evidence-based design principles addressing engagement Gap 3, derived from rigorous qualitative research; (2) methodological demonstration that qualitative, phenomenological approaches complement quantitative psychometric validation (RK) and mechanistic decoding (HD-EMG), producing actionable HCI insights; (3) bridge between novice musical learning and stroke rehabilitation (shared fine motor demands, sustained practice requirements); and (4) direct inform for integrating RK metrics and HD-EMG biofeedback into engaging, user-centered interfaces respecting lived experience, agency, and meaningfulness. By establishing mixed-methods approach, ReTouche ensures laboratory rigor translates into sustained clinical practice. **Detailed phenomenological analysis, methodological triangulation, and design implications are presented in Chapter 5.**

In summary, the three projects—RK, HD-EMG + AI, ReTouche—form a coherent, mutually reinforcing tripartite architecture addressing measurement, mechanism, and engagement gaps simultaneously. RK provides validated behavioral metrics, HD-EMG + AI reveals neuromuscular substrates, and ReTouche ensures tools are usable and engaging. Music unifies all three lenses as a precision testbed supporting rigorous measurement, auditory-motor coupling, and intrinsic motivation. This integration advances rehabilitation from descriptive assessment toward mechanistic understanding, from generic interventions toward personalized protocols, and from laboratory efficacy toward real-world effectiveness.

1.4 Research Questions, Hypotheses & Scope

1.4.1 Research Questions

The three fundamental gaps identified in Sections 1.1 and 1.4 (State of the Art) constitute the central research questions that structure this thesis:

Research Question 1 [Measurement]: Can music-based sensorimotor synchronization (SMS) paradigms, integrated with precision instrumentation (Dextrain Manipulandum, 0.01 N resolution, 1 kHz sampling), provide psychometrically validated, millisecond-resolution metrics of timing precision for fine manual dexterity in stroke rehabilitation?

Research Question 2 [Mechanism]: Can artificial intelligence (AI)-based regression models decode continuous, multi-degree-of-freedom (multi-DOF) finger forces from 224-channel high-density electromyography (HD-EMG) in real time with sufficient accuracy ($R^2 > 0.75$, latency <50 ms) and cross-user generalization to provide clinically interpretable neuromuscular readouts and mechanistic links between behavioral precision (RK metrics) and underlying control processes (co-contraction, individuation, motor unit recruitment)?

Research Question 3 [Engagement]: What evidence-based design principles—grounded in the lived experience of learning fine manual coordination in a musical context—are necessary to ensure

that precision measurement tools (RK) and mechanistic biofeedback systems (HD-EMG + AI) are not only scientifically rigorous but also usable, meaningful, and sustainable for long-term engagement in rehabilitation practice?

1.4.2 Hypotheses

Each research question is operationalized through specific, testable hypotheses:

Hypothesis Set 1: Psychometric Validation of RK (Addressing RQ1)

H1.1 [Reliability]: RK timing metrics (asynchrony variability, lag-1 autocorrelation, phase drift) will demonstrate excellent test-retest reliability, with Intraclass Correlation Coefficients (ICC) exceeding 0.75 for the primary metric (asynchrony variability) and 0.70 for secondary metrics, when assessed across two sessions separated by 7-14 days in both healthy participants ($N \geq 20$) and chronic stroke survivors ($N \geq 20$), following best-practice guidelines (Koo and Li, 2016).

H1.2 [Construct Validity - Convergent]: RK timing metrics will demonstrate moderate positive correlations (Pearson $r = 0.40$-0.70, $p < 0.05$) with established clinical measures of hand function (Fugl-Meyer Assessment Upper Extremity, Box and Block Test) and with Dextrain static force metrics (individuation index, force variability), indicating that timing precision measured by RK reflects underlying fine motor control capacity.

H1.3 [Construct Validity - Discriminant]: RK timing metrics will significantly discriminate between chronic stroke survivors and age-matched healthy controls, with stroke survivors exhibiting higher asynchrony variability (Cohen's $d > 0.80$), weaker error correction (lower magnitude lag-1 autocorrelation, $d > 0.60$), and greater phase drift ($d > 0.60$), demonstrating sensitivity to stroke-related sensorimotor deficits.

H1.4 [Experimental Sensitivity]: RK metrics will be sensitive to systematic auditory feedback manipulations predicted by sensorimotor synchronization (SMS) theory (Repp and Su, 2013; Repp, 2005): (a) Delayed Auditory Feedback (DAF) at 100-150 ms will increase asynchrony variability and reduce error correction magnitude compared to unperturbed feedback ($p < 0.05$); (b) P-center-aligned speech-like stimuli (Voks synthesizer) will reduce asynchrony variability by 15-25% compared to acoustically aligned stimuli ($p < 0.05$); (c) familiar, positively valenced musical stimuli will reduce asynchrony variability and enhance error correction compared to unfamiliar or neutral stimuli ($p < 0.05$).

Hypothesis Set 2: HD-EMG Decoding Performance and Mechanistic Linkage (Addressing RQ2)

H2.1 [Decoding Accuracy]: AI-based regression models (CNNs, LSTMs, CNN-LSTM hybrids) trained on 224-channel HD-EMG will achieve coefficient of determination $R^2 > 0.75$ (Pearson $r > 0.87$) for continuous prediction of multi-DOF finger forces during rhythmic tapping and force modulation tasks in healthy participants ($N \geq 15$), meeting the threshold for clinically meaningful interpretability (Muceli and Farina, 2012).

H2.2 [Real-Time Latency]: The optimized HD-EMG decoding pipeline will achieve end-to-end latency (EMG acquisition $\rightarrow$ force prediction output) of <50 ms, preserving the sense of agency required for effective real-time biofeedback (Farina et al., 2010).

H2.3 [Cross-User Generalization]: Models trained on pooled data from multiple participants will achieve cross-user $R^2 \geq 0.70$ when tested on held-out participants whose data were not used in training, demonstrating sufficient generalization to avoid the need for extensive per-user calibration in clinical settings.

H2.4 [Intrinsic Muscle Contribution]: Inclusion of 64 electrodes targeting intrinsic hand muscles (thenar, hypothenar, interossei) will improve finger individuation decoding accuracy by ≥ 5 percentage points in R^2 compared to models using only extrinsic (forearm) muscle signals, demonstrating unique, non-redundant information from intrinsic muscles critical for fine dexterity (Muceli and Farina, 2012).

H2.5 [Behavior-Mechanism Linkage]: Individuals exhibiting superior RK performance (lower asynchrony variability, stronger error correction) will demonstrate more selective HD-EMG activation patterns (lower co-contraction indices, higher individuation selectivity) during synchronized tapping tasks, as revealed by correlational analysis linking behavioral precision (RK metrics) to neuromuscular control quality (HD-EMG-derived features) ($r > 0.40$, $p < 0.05$).

Hypothesis Set 3: Engagement Design Principles from ReTouche (Addressing RQ3)

H3.1 [Interface Preference]: In the Comparative Structured Observation (CSO) study with counterbalanced conditions ($N = 18$ novice piano learners), participants will prefer the full ReTouche condition (visual projection mapping + haptic guidance + scaffolding features) over baseline (minimal feedback) and intermediate conditions (visual-only) on subjective ratings of engagement, perceived control, and satisfaction (Wilcoxon signed-rank test, $p < 0.05$), with at least 70% of participants ranking ReTouche highest.

H3.2 [Error Externalization]: Thematic analysis of autoethnography journals ($N = 3$, 2-week longitudinal practice) and CSO post-session interviews ($N = 18$) will reveal recurring themes indicating that embodied representations (replay, timeline visualizations, spatial projections) externalize errors and support strategic correction, with at least 60% of participants explicitly reporting instances where externalized feedback revealed errors they had not consciously detected during real-time performance.

H3.3 [Attentional Reorientation]: Participants will report (via post-session questionnaires and qualitative interviews) that ReTouche multimodal feedback shifted their attentional focus from task outcomes ("Did I play the right notes?") toward movement processes ("How smooth/well-timed were my finger movements?"), with process-oriented attention reported by $\geq 60\%$ of participants in the full ReTouche condition compared to $\leq 30\%$ in the baseline condition (Chi-square test, $p < 0.05$).

H3.4 [Agency and Sustained Engagement]: Autoethnography participants ($N = 3$) will report sustained engagement over the 2-week practice period (adherence $\geq 80\%$ of scheduled sessions) and will cite agency-supporting features (tempo control, segmentation, meaningful choices, immediate effects) as key motivators in at least 50% of journal entries, consistent with Self-Determination Theory predictions (Kirsh, 2013).

1.4.3 Scope and Limitations

While the three projects—RK, HD-EMG + AI, and ReTouche—make substantial contributions to their respective domains, it is essential to clarify the **scope and boundaries** of each contribution to ensure that claims remain rigorously grounded:

RK is a psychometric validation and measurement tool, not a clinical efficacy trial. RK establishes reliable, valid, sensitive timing precision metrics (addressing H1.1-H1.4) but does not constitute a randomized controlled trial (RCT) demonstrating therapeutic efficacy. The thesis does not claim that RK tasks themselves produce clinical improvement; rather, RK provides the outcome measure needed to evaluate rhythm-based interventions in future trials.

HD-EMG + AI is a proof-of-concept mechanistic exploration, not a deployment-ready system. The project demonstrates technical feasibility (H2.1-H2.4) and preliminary behavior-mechanism linkage (H2.5) but remains a research prototype requiring further validation (robustness, long-term stability, diverse stroke populations) before clinical deployment. The thesis establishes technical feasibility and mechanistic interpretability, not proven therapeutic benefit.

ReTouche provides qualitative insights and design principles, not quantitative efficacy evaluation. Qualitative methods (H3.1-H3.4) excel at revealing experiential depth and design affordances but do not quantify clinical efficacy. The thesis provides design knowledge and HCI framework to guide interface development and generate hypotheses for future mixed-methods evaluations integrating qualitative insights with quantitative outcomes (RK metrics, adherence tracking).

By explicitly stating these scope limitations, the thesis maintains a rigorous, measured tone that positions each project appropriately within the broader research trajectory—from measurement validation and mechanistic exploration to design principle derivation, with future clinical efficacy trials required to establish therapeutic impact.

General Discussion & Perspectives — Map

- **Integration**
  - Measure (RK) → Model (HD-EMG) → Engage (ReTouche): from outcomes to mechanisms to sustained use.
  - Music threads the triad: demands precision, reveals control, motivates practice (Zatorre, Chen, and Penhune, 2007; Chen, Penhune, and Zatorre, 2008).
- **Clinical perspectives**

  - RK metrics for assessment and tracking; HD-EMG biofeedback for quality of movement; ReTouche design levers for home adherence (Térémetz et al., 2015; Daly and Wolpaw, 2008; Langhorne, Bernhardt, and Kwakkel, 2011).

- **Methodological reflections**
  - Mixed methods: positivist quant (validity, ICC, generalization) + phenomenological depth (experience/meaning).
  - Real-time constraints: latency budgets for feedback; robust decoding under fatigue/noise (Englehart and Hudgins, 2003; Hahne et al., 2014).

- **Future directions**
  - Larger stroke samples; longitudinal/home studies; closed-loop RK+EMG with adaptive difficulty; principled interface design for long-term engagement (Braun Janzen et al., 2022).

Appendix A

Frequently Asked Questions

A.1 How do I change the colors of links?

The color of links can be changed to your liking using:
`\hypersetup{urlcolor=red}`, or
`\hypersetup{citecolor=green}`, or
`\hypersetup{allcolor=blue}`.
If you want to completely hide the links, you can use:
`\hypersetup{allcolors=.}`, or even better:
`\hypersetup{hidelinks}`.
If you want to have obvious links in the PDF but not the printed text, use:
`\hypersetup{colorlinks=false}`.

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